Regulatory Capture by Superstars: Lobbying, Innovation, and Industry Dynamics*

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Abstract

Firms lobby for two different things: transfers to themselves, such as tax provisions and government contracts, and the regulatory requirements that set the fixed cost of operating for every firm in their industry. We classify the universe of federal lobbying reports filed by public firms and trade associations, together with the bills they cite, into a closed taxonomy of 87 policy instruments, recording for each issue the instrument at stake, the change requested, and the firms it applies to. Firms agree on transfers, and nearly all ask for more, but they split on regulatory fixed costs: 56 percent of the firms and 38 percent of the trade associations that state a side ask for higher ones, and the largest firms ask most often. Participation in lobbying rises steeply with firm size on every front and is highly persistent. We build a dynamic general equilibrium model of oligopolistic competition and innovation in which superstar firms lobby for their own taxes and contracts and, against a trade association of small firms, for an industry-wide regulatory fixed cost set in an influence contest. Superstars favor higher fixed costs: leaders because marginal small firms exit and the leaders' profits rise, smaller superstars because the costs deter the entrants that would displace them. We estimate the model on firm dynamics, lobbying participation and outcomes, and the enactment of 2,656 regulatory proposals. Lobbying raises the regulatory fixed cost by 20 percent and costs 0.89 percent of lifetime consumption, mostly through lower output as industries become more concentrated and the competitive fringe shrinks, and a quarter through slower growth. Lobbying over regulation accounts for the entire cost; lobbying for tax cuts and contracts is nearly a pure transfer.

Keywords: lobbying, regulatory capture, innovation, industry dynamics, oligopoly, growth.
JEL Classification: D72, L13, L51, O31, O41.

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1 Introduction

Federal lobbying in the United States amounts to between four and five billion dollars a year, and Americans consistently rank the role of money in politics among the country's most serious problems. Yet what firms buy with this spending, and what it does to the economy, remains poorly understood. The canonical economic view treats lobbying as rent seeking: firms compete for transfers, and society loses the resources spent on the competition (Tullock, 1967; Krueger, 1974). A second tradition, going back to Stigler (1971), holds that the more consequential product of lobbying is regulation itself: incumbents shape the rules of their industry to raise the costs of rivals and keep entrants out (Salop and Scheffman, 1983). The two views have very different implications. A transfer redistributes surplus among firms and taxpayers. A regulatory requirement that every firm must meet changes the fixed cost of operating, and with it the number of firms an industry supports, the pressure on incumbents to innovate, and the rate at which new firms grow into large ones.

This paper measures what firms lobby for and quantifies the consequences of each kind of lobbying in a dynamic model of industry competition and innovation. We distinguish lobbying for *transfers to the firm* (its own tax burden and its own government contracts) from lobbying over *regulatory fixed costs*: licenses, product certification, recurring reporting, labeling, and privacy and cybersecurity rules, requirements that apply to every firm in an industry and do not scale with the firm's size. We call the latter *barrier lobbying*. The distinction matters because, unlike transfers, barriers are contested. A firm that asks for a lower tax rate asks for something every firm wants for itself; a firm that asks for a stricter compliance rule asks for something that raises its own costs, and does so only if the rule hurts its rivals more.

Our empirical contribution is a measure of the content of lobbying at the level of the policy instrument. The Lobbying Disclosure Act requires every registered lobbyist to report, each quarter, the client, the amount spent, and a free-text list of the issues lobbied, including the bills involved. We split the 2.5 million issues listed in the reports of Compustat firms (2008–2022) and of industry trade associations (2008–2018) into atomic requests, resolve every cited bill and public law against the congressional record, and assign each request to one instrument of a closed codebook of 87 federal policy instruments. The codebook is built from official inventories of each policy domain, from the Treasury's tax-expenditure list to the international classification of non-tariff measures, and each instrument is mapped in advance to the term of the firm's problem it moves. For each request, a language model records the instrument, the change requested relative to current law, whom the policy covers, and the position the filer takes; the economic content of the classification sits in a table written before any text is read.

Four facts emerge. First, firms lobby most on government demand (appropriations and procurement) and on regulatory fixed costs: lobbying firms lobby on a regulatory fixed cost in 46 percent of the quarters in which they file a report, and the largest 5 percent of firms in 77 percent. Second, firms agree on transfers and disagree on barriers. Between 95 and 100 percent of the firms that state a side on funding, contracts, grants, or credits ask for more, and 93 percent of those that state

a side on their profit tax ask for a lower one. On regulatory fixed costs, 56 percent of firms and 38 percent of trade associations ask for a *higher* barrier, and the share rises with size: 42 percent of firms below the 80th percentile of sales, 64 percent of the largest 5 percent. Third, lobbying is a large-firm activity on every front: within an industry and year, doubling a firm's sales raises the probability that it lobbies on barriers by 1.9 percentage points, against an average of 6 percent. Fourth, lobbying is persistent: nine in ten firms that lobby on a front in one quarter are still on it the next.

To interpret these facts and quantify their consequences, we build a dynamic general equilibrium model of oligopolistic competition and innovation in the tradition of step-by-step growth models (Aghion et al., 2001), extending the quality-ladder framework of Cavenaile et al. (2026). Each industry has up to four strategic *superstar* firms that compete à la Cournot and innovate to climb a quality ladder, and an actual competitive fringe: thousands of atomistic firms that differ in their fixed costs of operating, decide every instant whether to produce, and occasionally break through into the superstar set. Superstars lobby on three fronts. They lobby for a lower own profit tax and for more own procurement contracts, and they lobby over the industry's regulatory fixed cost, which is decided by an influence contest in the spirit of Becker (1983) between the superstars and a trade association that coordinates the fringe, subject to an Olson-style cost of organizing many members (Olson, 1965). Each decision to participate in lobbying carries a fixed cost that is redrawn at random times, which generates the persistence of lobbying and its steep gradient in firm size. The equilibrium is Markov perfect: firms choose innovation, lobbying intensities, and participation knowing how regulation feeds back into the fringe, entry, and their own future position.

The model's central object is the marginal value of regulation to an incumbent superstar. It has three parts: the firm's own compliance cost, the Cournot gain as marginal fringe firms stop producing and the firm's share rises, and the value of fewer fringe breakthroughs that would push the firm down the ranks or out of the superstar set. At our estimates, the gain exceeds the cost for every superstar, but for different reasons. For the leader, the stake is mostly static: 58 percent of its gain comes from the profits it takes over from exiting fringe firms. For the fourth superstar, the stake is mostly dynamic: 75 percent of its gain comes from deterring entry, because an entrant that joins a full industry displaces the smallest incumbent. An entrant joining a full top four lowers the leader's profit by 6 percent and the fourth superstar's by 80 percent. The fringe, by contrast, bears the compliance cost without the protection, and its association lobbies against the barrier.

We estimate the model by the simulated method of moments in three blocks that are solved jointly. The technology block relies mainly on firm dynamics rather than on patent regressions: the distribution of sales gaps among the four largest firms, the rate and location of entry into the top four, how gaps between the same two firms evolve, markups, growth, and the entry, exit, and losses of small firms from Census and Compustat data. The tax and procurement block uses the outcomes that lobbying buys, which the data reveal directly: the change in a firm's tax rate and in its federal contracts after it lobbies, together with who lobbies and how persistently. The regulation block faces the difficulty that the regulatory fixed cost is not observed by industry and year. We

identify the contest from the bills that set it: we simulate each of the 2,656 proposals that raised or lowered regulatory fixed costs in 2008–2022 with every industry that lobbied it, in the model, and run the same enactment regressions on the real and the simulated proposals. A global search over 8,192 points of a 37-dimensional parameter space, each screened on its technology fit and then solved to a certified equilibrium, selects the estimate.

Our main quantitative result is that lobbying raises the regulatory fixed cost by 20 percent relative to an economy without lobbying. Superstars win the contest: their stakes are large and concentrated in a few firms, while the association must coordinate thousands of members at a cost that rises steeply with membership. The higher barrier shrinks the operating fringe by 3.5 percent and reduces the rate at which fringe firms break into the top four. Industries shift toward states with fewer superstars and larger leads, the superstars that remain take a larger share of their markets, and growth falls by 0.9 basis points as fewer industries have a close race at the top. Lobbying costs the representative household 0.89 percent of lifetime consumption. Most of the cost is static: 57 percent is lower output from the more concentrated industry structure and the smaller fringe, 17 percent is the compliance and lobbying resources net of the fringe fixed costs saved, and 25 percent is slower growth.

Counterfactuals that ban one kind of lobbying at a time show that barrier lobbying accounts for the entire welfare cost. With lobbying allowed only over regulation, the cost is 0.88 percent of consumption; with lobbying allowed only for taxes and contracts, it is 0.0003 percent. Tax and procurement lobbying are nearly pure transfers in the model: the household that owns the firms also pays the labor tax that finances the lost revenue, the contracts deliver goods worth their cost, and the small distortions of labor supply in opposite directions nearly cancel. This finding inverts the emphasis of the rent-seeking view and of public debate, which focus on firm-specific tax breaks and contracts: the costly part of lobbying is the least visible one, the contest over the rules that set the cost of competing.

The sign of the welfare effect is not built into the model. If the trade association won the contest, lobbying would lower regulatory fixed costs and raise welfare. Among the parameter values that fit the moments other than the bills best, the single best-fitting one is such a world: the barrier falls by 36 percent and lobbying raises welfare by 0.35 percent. The bill data reject it decisively. In that world superstars lobby to lower compliance costs, so lobbying superstars would raise the enactment of proposals that lower them by 14 percentage points per log point and barely help proposals that raise them (1.6 points); in the data the first effect is -0.4 points and the second 5.1. Among the ten parameter values that fit all moments best, every one has lobbying raising the barrier, by between 6 and 150 percent, and every one implies a welfare loss.

Related literature. The paper contributes to three literatures. The first studies lobbying and the political economy of regulation. Theories of regulatory capture (Stigler, 1971; Peltzman, 1976) and of influence contests (Becker, 1983; Grossman and Helpman, 1994; Becker and Mulligan, 2003) established how organized interests shape policy; a large empirical literature measures who lobbies

and what it buys (Bombardini, 2008; Richter et al., 2009; Bertrand et al., 2014; Kerr et al., 2014; Kang, 2016; Kim, 2017, 2018; Blanes i Vidal et al., 2012), and firms' choice between lobbying alone and through associations (Bombardini and Trebbi, 2012). Quantitative models have studied lobbying by atomistic firms for tax benefits or sales (Arayavechkit et al., 2018; Hossein Dad and Nguyen, 2025; Huneeus and Kim, 2024) and the political connections of incumbents (Akcigit et al., 2023). We contribute a measure of the content of lobbying at the level of the policy instrument, the change requested, and the side taken, and a framework in which large strategic firms and an association of small firms lobby non-cooperatively over both firm-specific and industry-wide policies in a dynamic economy. Closest to our empirical exercise, Macedoni and Weinberger (2024) classify lobbying by topic and show that firms with high sales relative to capital support increases in fixed costs; we measure the instrument and the direction of each request and estimate the structural consequences.

The second literature studies how regulation shapes entry, concentration, and growth. Entry regulation is costly (Djankov et al., 2002), regulation weighs disproportionately on small firms and new businesses (Bailey and Thomas, 2017; Singla, 2023), and size-contingent regulation distorts the firm size distribution and innovation (Garicano et al., 2016; Aghion et al., 2023). The rise in US concentration and the decline in business dynamism have been attributed in part to barriers to entry and to lobbying by incumbents (Gutiérrez and Philippon, 2019; Covarrubias et al., 2020; Philippon, 2019; Zingales, 2017; Akcigit and Ates, 2021, 2023). We provide a model in which the regulatory fixed cost of an industry is an equilibrium outcome of a contest between incumbents and small firms, and quantify how much of the regulatory burden lobbying adds, through which margins it reduces welfare, and what share of the cost each kind of lobbying causes. The measurement of compliance costs by Trebbi et al. (2023) and of firms' regulatory exposure by Kalmenovitz (2023) disciplines the level of the burden.

The third literature studies markups, innovation, and growth in models with strategic firms. Step-by-step innovation models (Aghion et al., 2001, 2005) and their quantitative descendants (Klette and Kortum, 2004; Akcigit and Kerr, 2018; Acemoglu et al., 2018; Peters, 2020; Liu et al., 2022; De Ridder, 2024) study how competition shapes innovation; Cavenaile et al. (2026) combine dynamic oligopoly with a competitive fringe to quantify the welfare cost of markups. Dynamic oligopoly models follow Ericson and Pakes (1995) and Doraszelski and Pakes (2007); we keep the state space tractable with analogy-based beliefs about rivals' tax and procurement statuses (Jehiel, 2005), in the spirit of approximations that replace rivals' detailed states with summary statistics (Weintraub et al., 2008; Benkard et al., 2015; Fershtman and Pakes, 2012). We add endogenous lobbying and a political equilibrium to this class of models, and find that the welfare cost of capture works mainly through industry composition: the number of superstars, the size of their leads, and the mass of competitive firms, rather than through the innovation rate of a given firm.

The paper is organized as follows. Section 2 describes the data and the classification. Section 3 presents the facts. Section 4 develops the model. Section 5 describes the estimation and the fit. Section 6 reports the political equilibrium, the effects of lobbying, its welfare cost, and the

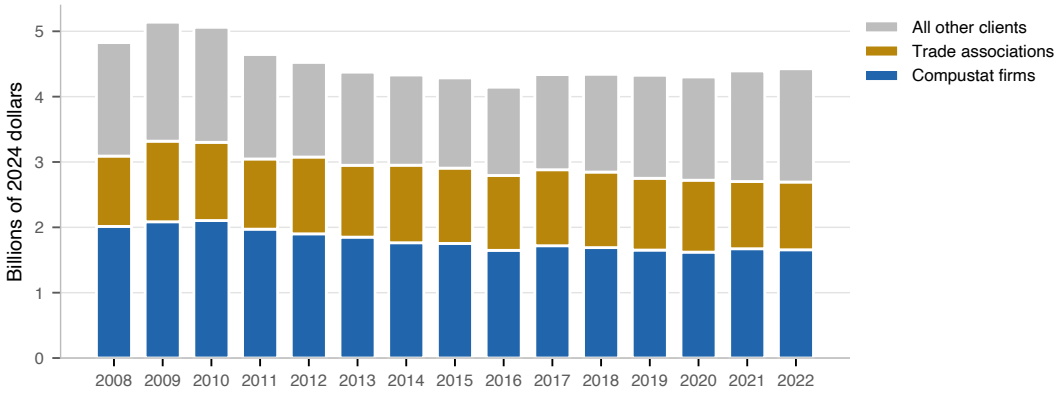


Figure 1. Who Lobbies: Compustat Firms and Trade Associations

Notes: Federal lobbying spending by type of client, billions of 2024 dollars, 2008–2022. Trade associations are clients matched to an IRS 501(c)(6) organization.

counterfactuals. Section 7 concludes.

2 Data and Measurement

This section describes the lobbying reports, the classification of their content into policy instruments, and the firm, industry, and legislative data we combine them with. Section 3 presents the facts.

2.1 Lobbying reports

The Lobbying Disclosure Act of 1995 requires every registered lobbyist to file a report (form LD-2) for each client every quarter. A report names the client and the lobbying firm (or the client's in-house lobbyists), states the amount spent, and lists the issues lobbied: for each general issue area, a free-text description of the specific issues, often with the bills, public laws, and regulations involved. We use every report filed between 2008 and 2022, obtained from the Senate Office of Public Records and cross-validated against LobbyView (Kim, 2018) and OpenSecrets (Center for Responsive Politics, 2024).

We match clients to two kinds of organizations. Public firms are matched to Compustat North America, using the firms' names, their subsidiaries as listed in their SEC filings, and their merger histories, so that a subsidiary's lobbying is attributed to its parent. Trade associations are matched to the IRS register of business leagues (501(c)(6) organizations) and assigned to the industries they represent. Together, Compustat firms and industry trade associations account for 65 percent of all lobbying dollars, 36 percent of the reports, and 52 percent of the issues listed in 2008–2022 (Figure 1). Our firm sample is every report of a Compustat firm in 2008–2022; our association sample is every report of an industry trade association in 2008–2018.

2.2 From text to policy instruments

The issue text is the central object of the measurement. A single issue block often bundles several requests on different policies, cites bills by number, abbreviation, or title, and states the client's position only occasionally. We process it in four steps.

Splitting. Each issue block is split into atomic requests, each concerning a single policy. Deterministic pattern rules handle unambiguous structure, such as bill-led segments, enumerated lists, and blank-line groups; blocks whose boundaries require judgment are split by a language model, under the constraint that the pieces reproduce the original text verbatim. Appendix H gives examples.

Bills. Every cited bill and public law is extracted, resolved against the congressional record, assigned to its Congress, and corrected for mistyped or non-existent numbers by matching titles within a window of Congresses. A bill enters the analysis through its provisions: titles, sections, and the bullets of its official summary are classified in the same way as issue text, so that citing an omnibus bill does not count as lobbying on everything it contains.

Instruments. Each request is assigned to one instrument of a closed codebook of 87 federal policy instruments. The codebook is built from the official inventories of each policy domain: the Treasury tax-expenditure list, the taxonomy of regulatory forms of Xie and Perez (2018), the Harmonized Tariff Schedule and the UNCTAD classification of non-tariff measures, the federal Assistance Listings and the Federal Acquisition Regulation, and the patent, exclusivity, and immigration statutes, with the tools-of-government catalog of Salamon (2002) as a check on the inventory as a whole. Each instrument is defined by the statutory or regulatory language that identifies it, and is mapped, by a table fixed before any text is read, to the term of the firm's problem it moves, which we call its *wedge*. Table 1 lists the ten wedges. The codebook is organized by what an instrument does rather than by whom it reaches: a research credit written for one firm and one available to the whole economy are the same instrument, distinguished by the eligibility conditions read from the text. Appendix I describes the codebook and the crosswalk.

Content of each request. Alongside the instrument, the classifier records the change requested relative to current law (introduce, raise, lower, extend, or repeal, with the magnitude when the text states it), whom the policy covers (industry, geography, product, legal form, and thresholds on firm characteristics), and the position the filer takes (support, oppose, amend, monitor, or unstated). The codebook fixes, for each instrument, the sign of its typical payoff to a covered firm and its shape (proportional or quasi-fixed). Two quantities are then derived mechanically. The *direct payoff* of a request is the direction of the change times the instrument's sign: creating a credit is favorable, creating an excise is not. The *side* the filer takes is its stance times the direction of the change: supporting a bill that raises a compliance requirement and opposing the repeal of one are the same

Table 1. Wedges of the Firm's Problem and the Instruments That Move Them

Wedge	Instruments (examples)	Number
Demand, s^d	appropriations and earmarks, procurement contracts, buyer purchase subsidies, tariffs and quotas, antidumping duties	14
Regulatory fixed cost, R_j	licenses and charters, product certification, pre-market approval, facility permits, recurring reporting, labeling, data privacy and cybersecurity, liability rules	18
Output, s^y	per-unit production credits, excise taxes, emission charges, performance standards with per-unit costs, quotas	17
Profit tax, b_T	the corporate income tax, deductions, reduced rates on classes of income, foreign income, losses	7
Innovation cost, s^r	research credits and expensing, research grants, patents, product exclusivity	5
Capital cost, s^k	investment credits, accelerated depreciation, tax-exempt bonds, federal lending	6
Labor cost, s^w	hiring credits, payroll taxes, wage floors, worker protections, visas	8
Set prices, s^p	government payment rates, mandatory discounts, price ceilings and floors, rate regulation	5
Competition, $A(\Theta)$	merger control, platform regulation, antitrust exemptions, mandated access	4
Lump-sum transfers, s^i	grants and direct payments, regulatory user fees	2
Place-based bundles	enterprise zones, decomposed into their components	1

Notes: The codebook's 87 instruments by the wedge of the firm's problem they move. A regulatory fixed cost is a requirement whose cost of compliance does not scale with the firm's output; requirements whose cost scales with output are output wedges. Appendix I lists the instruments.

political act, and both place the filer on the side of a higher regulatory fixed cost.

Measurement rules. A firm lobbies on an instrument in a quarter if one of its reports names the instrument, or a bill title or section that contains it, at high or medium confidence; or cites a regulation that contains it; or cites a bill that moves only that wedge. A bare citation of a bill that moves several wedges counts for none of them, since citing a defense authorization would otherwise make a firm active on procurement, export controls, and cybersecurity alike. A firm-year's side on an instrument is the direction most of its requests on the instrument point to; on regulation, a stance on a bill that changes only regulation also counts. Dollars are whole report amounts and are never split across a report's issues, so a report counts for every front it touches. Table 2 reports the coverage of the classification.

Validation. An independent classification of 2,000 randomly drawn issues by a second, larger language model agrees with ours on 84 percent of them, and the sides we record agree with the hand-coded positions of MapLight on 96 percent of the firm–bill pairs both cover. Classifying the same issues twice gives the same instruments in 94 percent of cases and the same side in 99 percent. A hand-coded validation sample, which will also be used to debias the estimates following Carlson and Dell (2025), is in progress.

Table 2. Classification: Coverage and Confidence

	Compustat firms 2008–2022	Industry associations 2008–2018
Reports with issue text	209,410	75,291
Issues listed in the reports	1,604,255	901,044
matched to a policy instrument	46%	46%
bare citation of a bill or rule	31%	37%
no policy instrument	10%	7%
unclear (e.g., "climate change", "tax reform")	13%	10%
Confidence of the matched issues:		
high	52%	46%
medium	37%	41%
low (dropped)	10%	12%

Notes: Bare citations enter through the provisions of the cited bill or rule (Section 2.2). Shares of the most lobbied instruments move by at most 4 percentage points when low-confidence requests are kept.

2.3 Firms, industries, and legislation

Firms and industries. Firm data come from Compustat for 2008–2022 (US-incorporated firms reporting in US dollars, excluding finance and insurance). Industries are four-digit NAICS codes, and we keep industry-years with at least six Compustat firms. We call a firm a *superstar* if it is among the four largest firms of its industry-year by sales and its sales are at least 10 percent of the leader's; every other firm belongs to the industry's *fringe*. Census data provide the industry concentration of the four largest firms (Economic Census), firm entry and exit (Business Dynamics Statistics), and business applications and the share that become employers (Business Formation Statistics). Aggregate growth is the growth of real GDP per capita; R&D intensity is business-funded R&D over GDP (National Science Foundation); the labor share is the corporate labor share of Karabarbounis and Neiman (2014), extended with the OECD sector accounts.

Taxes and contracts. A firm's tax rate is its book tax expense over pretax income in profitable years. Federal procurement contracts are prime awards from the Federal Procurement Data System, matched to Compustat parents with the same name, subsidiary, and merger resolution as the lobbying reports; joint ventures are split among their owners. The matched contracts cover 54 to 59 percent of federal contract dollars.

Legislation. Bills and their histories come from Congress.gov. We group the bills of one Congress that are the same legislation into a *proposal*, and use the provision-level classification to identify proposals that raise or lower regulatory fixed costs: in 2008–2022, 2,020 proposals raise them, of which 118 were enacted, and 636 lower them, of which 46 were enacted.

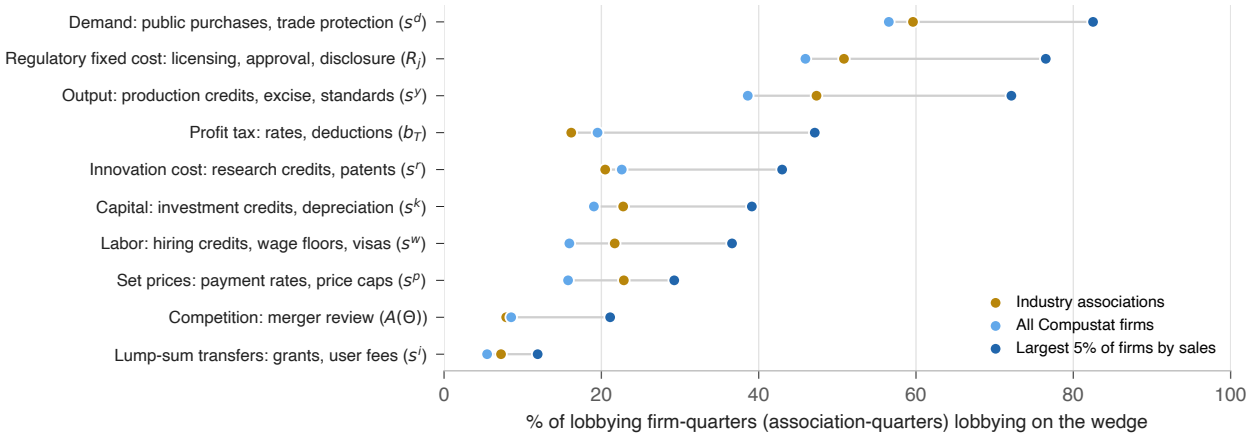


Figure 2. Lobbying by Wedge of the Firm's Problem

Notes: Share of lobbying firm-quarters (association-quarters) in which the client lobbies on the wedge. 71,433 Compustat firm-quarters, of which 14,127 of the largest 5 percent of firm-years by sales within the year; 36,939 association-quarters. Compustat firms 2008–2022, associations 2008–2018.

3 What Firms Lobby For

We document four facts about the content and the structure of lobbying. Together they motivate the model of Section 4: lobbying is concentrated on two kinds of policy, firms agree on one and disagree on the other, and the decision to lobby is a persistent, large-firm choice.

3.1 Fact 1: Firms lobby most on government demand and on regulatory fixed costs

Figure 2 reports, for each wedge, the share of the quarters with a lobbying report in which the firm (or association) lobbies on it. Two wedges dominate. Lobbying firms lobby on government demand (appropriations, procurement, and buyer subsidies) in 57 percent of their quarters, and on regulatory fixed costs in 46 percent; for trade associations the shares are 60 and 51 percent. The profit tax, which dominates public discussion of corporate lobbying, appears in about a fifth of the quarters. The largest 5 percent of firms lobby on every wedge more often, on government demand in 83 percent of their quarters and on regulatory fixed costs in 77 percent.

At the level of instruments, program funding and appropriations lead for everyone (29 percent of firm-quarters), followed by technology and performance standards (19 percent), data privacy and cybersecurity, the most lobbied regulatory fixed cost (19 percent), procurement contracts (18 percent), buyer subsidies and purchase mandates (18 percent), and recurring reporting and record-keeping duties (16 percent). Associations lobby more than firms on standards, reporting duties, and payment rates, and less on procurement (9 versus 18 percent). Each industry lobbies on what it sells to the government and on what its regulator controls, and a regulatory fixed cost is among the three most lobbied instruments in five of the six largest lobbying sectors: transaction rules in health care services and in finance, pre-market approval in pharmaceuticals and devices, privacy in

Table 3. What Each Sector Lobbies on Most

	Most common	Second	Third
Aerospace and defense	Procurement (s^d)	Appropriations (s^d)	R&D grants (s^r)
Health care services	Payment rates (s^p)	Appropriations (s^d)	Transaction rules (R_j)
Pharma and devices	Appropriations (s^d)	Payment rates (s^p)	Pre-market approval (R_j)
Technology	Appropriations (s^d)	Procurement (s^d)	Privacy (R_j)
Finance and insurance	Transaction rules (R_j)	Buyer subsidies (s^d)	Privacy (R_j)
Energy and utilities	Facility permits (R_j)	Technology standards (s^y)	Profit tax (b_T)

Notes: The three instruments lobbied in the largest share of the sector's lobbying firm-quarters, Compustat firms, 2008–2022.

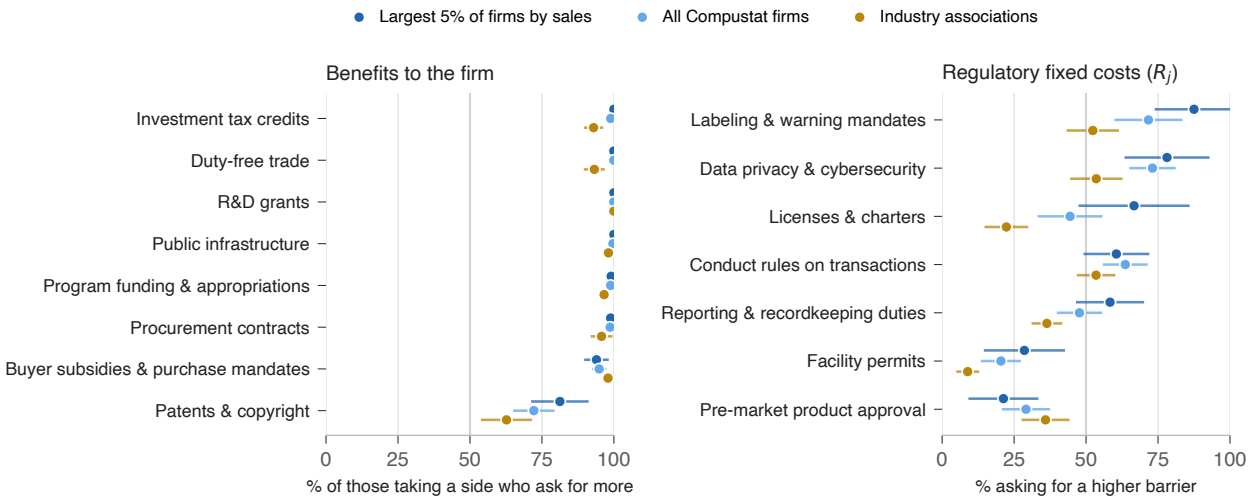


Figure 3. The Side Firms Take: Benefits and Regulatory Fixed Costs

Notes: Left: share of those taking a side who ask for more of the benefit. Right: share asking for a higher regulatory fixed cost. A firm-year's side is the direction most of its requests on the instrument point to; on regulation, a stance on a bill that changes only regulation also counts. Instruments with at least 50 firm-years taking a side; 95 percent confidence intervals.

technology, and facility permits in energy and utilities (Table 3).

3.2 Fact 2: Firms agree on transfers and disagree on barriers

Figure 3 reports, for each instrument with at least 50 firm-years taking a side, the share of those taking a side who ask for more of it. On instruments that benefit the firm directly, firms speak with one voice: between 95 and 100 percent of those taking a side ask for more funding, contracts, grants, and credits (72 percent on patents and copyright, an instrument that favors firms holding intellectual property and burdens those using it), 93 percent of those taking a side on the profit tax ask for a lower one, and 99 percent ask for more contracts. On regulatory fixed costs, firms split. Depending on the instrument, between 20 and 73 percent of the firms that take a side ask for a *higher* barrier; pooling the barrier instruments, 56 percent of firms and 38 percent of trade associations do.

The firms that ask for higher barriers are disproportionately large. The share asking for a

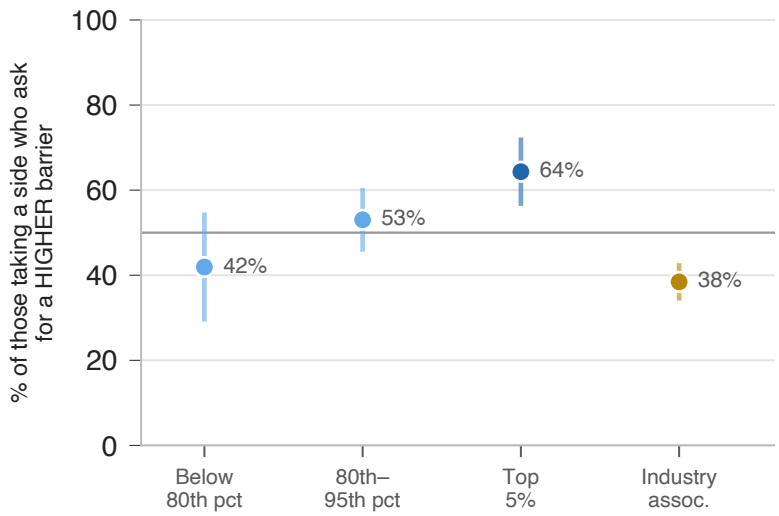


Figure 4. Share Asking for a Higher Barrier, by Firm Size

Notes: Share of firm-years (association-years) taking a side on regulatory fixed costs that ask for a higher barrier, by the firm's sales percentile within the year. 417 firm-years and 614 association-years take a side. A stance on a bill counts only when the bill changes nothing but regulation. 95 percent confidence intervals.

higher barrier is 42 percent among firms below the 80th percentile of sales within the year, 53 percent between the 80th and 95th percentiles, and 64 percent among the largest 5 percent; trade associations, which represent smaller firms, are the least likely to ask for higher barriers (Figure 4). On each of four main barriers, the largest 5 percent ask for a higher barrier more often than other firms: labeling (88 versus 60 percent), licenses (67 versus 35), reporting duties (58 versus 39), and privacy (78 versus 73). The gradient is imprecisely estimated, however, because firms rarely state a side: 417 firm-years take a side on barriers. Pooling the barrier instruments, the largest 5 percent ask for a higher barrier 14 percentage points more often than other firms (standard error 8), and across cutoffs for "the largest firms" from the top 1 to the top 20 percent the difference ranges from 3 to 16 points. Within an industry-year, only 8 to 30 industry-years have firms taking a side on both sides of a size cutoff, too few to estimate the gradient. We therefore use the overall share of sides in estimation and treat the size gradient as an untargeted pattern.

A natural concern is that firms state a side selectively, for instance only when they oppose a rule. The data indicate instead that whether a report states a direction is a habit of the lobbying firm that writes it. The median lobbying firm states a direction on 0.5 percent of the issues it reports, and 41 percent never do. A lobbying firm's rate of stating a direction with half of its clients predicts its rate with the other half (correlation 0.66), and a client's reports state a direction more often when its lobbying firm does so for its other clients (coefficient 0.48, standard error 0.11). On bills on which MapLight records a firm as supporting, its reports state a side 7.6 percent of the time; on bills it opposes, 7.5 percent. Stated sides are thus a sample selected on the lobbyist rather than on the side, and their composition is informative about the composition of all sides.

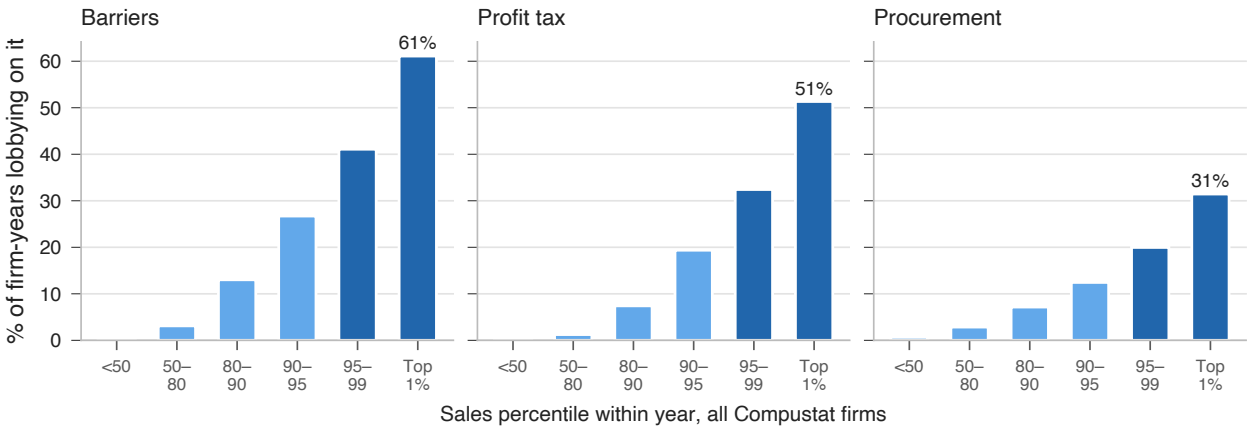


Figure 5. Participation in Lobbying by Firm Size

Notes: Share of firm-years lobbying on each front, by the firm's sales percentile within the year. All 110,836 Compustat firm-years, 2008–2022.

Table 4. Superstars' Participation and Persistence

	Lobbying in a quarter, by rank				Next quarter	
Front	Leader	#2	#3	#4	Stay	Join
Barriers	0.365	0.232	0.185	0.143	0.924	0.026
Profit tax	0.273	0.168	0.126	0.121	0.910	0.021
Procurement	0.192	0.111	0.079	0.075	0.885	0.015

Notes: Superstars are the four largest firms of a four-digit NAICS industry-year with sales of at least 10 percent of the leader's. A firm lobbies on a front in a quarter if any of its reports in the quarter lobbies on it; monitoring counts. Stay: probability of lobbying on the front next quarter given lobbying this quarter; join: given not lobbying.

3.3 Fact 3: Lobbying rises steeply with firm size on every front

We now focus on three fronts that the model will carry: the firm's own profit tax, its own federal procurement contracts, and *barriers*, the regulatory fixed costs that do not grow with the firm's size (licenses, product certification and inspection, recurring reporting, labeling, privacy and cybersecurity, and services-trade and data-flow rules). Reports that include these fronts carry 64 percent of firms' lobbying dollars, and the fronts span both kinds of policy: tax and procurement are one-sided, barriers two-sided. Participation on each front rises steeply with size (Figure 5). Below the median of sales almost no firm lobbies; among the largest 1 percent of firm-years, 61 percent lobby on barriers, 51 percent on the profit tax, and 31 percent on procurement. Within an industry and year, each log point of sales raises the probability of lobbying on barriers by 2.7 percentage points, on the profit tax by 1.9, and on procurement by 1.5 (standard errors 0.1; 110,379 firm-years), against an average of 6 percent. Among superstars the gradient continues by rank: in a given quarter, 37 percent of industry leaders lobby on barriers, against 23, 18, and 14 percent of the second through fourth firms (Table 4).

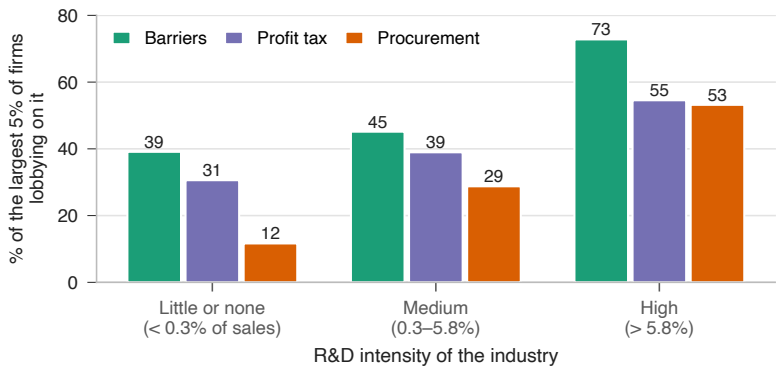


Figure 6. Barrier Lobbying by the Largest Firms and Industry R&D Intensity

Notes: Share of the largest 5 percent of firm-years by sales that lobby on each front, by the R&D intensity of their four-digit NAICS industry (R&D over sales in the year), split at the median and the 75th percentile of firm-years. 3,045, 1,845, and 658 firm-years.

3.4 Fact 4: Lobbying is persistent

Nine in ten firms that lobby on a front in one quarter lobby on it in the next: 91 percent on barriers, 88 percent on the profit tax and on procurement among all Compustat firms, and 94, 91, and 89 percent among the largest 5 percent. Few firms join in a given quarter (0.4 to 0.5 percent of all firms, 2.4 to 4.2 percent of the largest). Trade associations are as persistent (91 percent on barriers). Lobbying thus looks like a costly organizational commitment, made at infrequent times and maintained, rather than a flow chosen anew each quarter.

3.5 Associations and the R&D intensity of industries

Two further patterns guide the model. First, trade associations and superstars lobby side by side. In a given industry-quarter, the association lobbies on barriers without any superstar in 27 percent of the cases, superstars without the association in 15 percent, both in 33 percent, and neither in 25 percent; the association is active in 57 percent of quarters, and when it is, superstars are 2.5 percentage points more likely to lobby. Second, barrier lobbying is concentrated in innovative industries. In industries in the top quartile of R&D intensity, almost three in four of the largest 5 percent of firms lobby on barriers, against 39 percent in industries below the median; the other firms lobby on barriers at a rate of 4 percent in every group (Figure 6). This is consistent with the mechanism of the model, in which the value of a barrier to an incumbent rises with the threat of entry through innovation.

3.6 From facts to model

The facts shape the model in four ways. Because tax and procurement lobbying is one-sided and firm-specific, the model treats it as each firm's own instrument, with an outcome observed in the data: the firm's tax rate and its contracts. Because lobbying over barriers is two-sided and the rule applies to every firm in an industry, the model determines the regulatory fixed cost in

a contest between firms with opposing stakes, and the stake of each firm, and hence its side, is an equilibrium object. Because lobbying is concentrated among the largest firms while small firms lobby through associations, each industry has a few strategic superstars that lobby individually and a competitive fringe that lobbies only through its association. And because participation is persistent and rises with size, the decision to lobby carries a fixed cost that is revised only at infrequent times. The model's moments are computed exactly as their data counterparts: participation by rank, its persistence, the share of sides, the association's activity, and the outcomes of tax and procurement lobbying.

4 Model

We extend the dynamic quality-ladder oligopoly framework of Cavenaile et al. (2026) in two directions. First, each industry has an actual competitive fringe: thousands of small firms with heterogeneous fixed costs of operating, which decide every instant whether to produce and occasionally break through into the superstar set. Second, firms lobby on the three fronts of Section 3: a cut in their own profit tax (T), their own procurement contracts (S), and an industry-wide regulatory fixed cost (R) decided by a contest in which superstars and a trade association of fringe firms can take opposite sides. We first present the environment, then the decisions of each agent, the equilibrium, and welfare.

4.1 Environment

Time is continuous, indexed by $t \in \mathbb{R}_+$. There is a representative household, a unit mass of industries indexed by $j \in [0,1]$, a unit mass of risk-neutral entrepreneurs, and a government.

Preferences. The preferences of the representative household are given by:

$$U = \int_0^\infty e^{-\rho t} \left[\ln C_t \; - \; \xi \, \frac{L_t^{\,1+1/\epsilon_L}}{1+1/\epsilon_L} \right] dt, \qquad C_t \; = \; C_t^p + G_t, \tag{1}$$

where C_t is total consumption: private consumption C_t^p, chosen by the household, plus government purchases of the final good G_t, which the household takes as given. L_t is labor supply and ϵ_L the Frisch elasticity of labor supply.[1]

Final good. The final good Y_t is produced competitively from a unit mass of industry outputs:

$$\ln Y_t \; = \; \int_0^1 \ln(y_{jt})\, dj. \tag{2}$$

[1] Government purchases are perfect substitutes for private consumption. As a robustness check, purchases can be valued at $v < 1$, with $C_t = C_t^p + v\, G_t$; the equilibrium does not depend on v, only welfare does.

We normalize the final-good price to one, and the Cobb–Douglas aggregation implies that each industry receives an equal expenditure share.

Industry structure. Each industry j at time t contains $N_{jt} \leq \bar{N}$ *superstar* firms competing in quantities and producing differentiated varieties, with $\bar{N} = 4$. Additionally, each industry has a mass m_t of atomistic *price-taking fringe* firms. Industry output aggregates varieties via constant elasticity of substitution:

$$y_{jt} \;=\; \left(\sum_{i=1}^{N_{jt}} y_{ijt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{cjt}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \qquad \eta > 1, \tag{3}$$

where $\tilde{y}_{cjt} = \int_0^{m_t} y_{ckjt}\, dk$ is aggregate fringe output.

Technology. All firms produce output using a technology with decreasing returns to scale in labor:

$$y_{ijt} \;=\; q_{ijt}\, l_{ijt}^{\alpha}, \qquad \alpha \in (0,1). \tag{4}$$

Superstars' productivities follow a quality ladder. Let $n_{ijt} \in \{0, 1, \ldots, \bar{n}\}$ denote the gap of superstar i behind the industry leader (the leader has a zero gap). Then

$$q_{ijt} \;=\; q_{jt}^{\text{leader}} \cdot (1+\lambda)^{-n_{ijt}}, \qquad \lambda > 0, \tag{5}$$

where q_{jt}^{leader} is the frontier productivity in industry j at time t. Fringe firms do not climb the quality ladder themselves; their productivity follows the industry's superstars. All fringe firms in industry j share the productivity

$$q_{cjt} \;=\; \zeta \Big(\prod_{k=1}^{\bar{N}} q_{kjt} \Big)^{1/\bar{N}}, \tag{6}$$

that is, ζ times the geometric mean of the qualities of the $\bar{N}$ superstar seats, where a seat without a superstar counts at the lowest quality a superstar can have, $q_{jt}^{\text{leader}}(1+\lambda)^{-\bar{n}}$. The fringe thus draws on the technology of all incumbents, not only the leader's: when every seat is filled, a step by any superstar raises fringe productivity by the factor $(1+\lambda)^{1/\bar{N}}$, so the innovating superstar gains $(\bar{N}-1)/\bar{N}$ of a step on the fringe, and an industry's fringe is more productive when its superstars are close to one another. The parameter ζ sets the fringe's productivity level, and with it the fringe's share of industry sales.

Fixed costs and firm heterogeneity. Every active firm pays two kinds of fixed costs. The first is regulatory compliance: each active fringe firm pays $R_{jt}Y_t$, and each superstar pays $\kappa_c R_{jt} Y_t$, where $\kappa_c \geq 1$ is the superstars' compliance multiple. The level R_{jt} is set by lobbying (Section 4.3). The second is an idiosyncratic operating cost $\varepsilon_k Y_t$, paid by fringe firms only. The draw ε_k is made at

the firm's birth and kept for its life:

$$\varepsilon_k \;\sim\; \text{LogN}(\mu_\varepsilon, \sigma^2), \qquad \mu_\varepsilon = \ln \bar{\varepsilon} - \sigma^2/2, \tag{7}$$

so that $\mathbb{E}[\varepsilon_k] = \bar{\varepsilon}$. The cumulative distribution and density of ε are denoted $F(\cdot)$ and $f(\cdot)$. A fringe firm can stop and restart producing at no cost. Every instant it operates if and only if $\varepsilon_k \leq \varepsilon^*_{jt}$. We call the operating firms *active*; their mass is $m^{\text{a}}_{jt} = m_t\, F(\varepsilon^*_{jt})$. The draw ε_k is the only source of heterogeneity among fringe firms, and it generates the *selection margin* that we exploit throughout: a higher R_{jt} lowers ε^*_{jt}, and the marginal fringe firms stop producing. Heterogeneous fringe fixed costs and a positive R_{jt} are the central structural deviation from Cavenaile et al. (2026): they create the margin through which regulation, and lobbying over it, operates.

Entry and exit. Entry into the fringe is governed by a unit mass of entrepreneurs who pay $\psi e_t^2 Y_t$ to generate a Poisson rate e_t of new fringe firms. New fringe firms draw ε_k from (7) and are randomly allocated across industries, implying $m_{jt} = m_t$ by ex-ante symmetry. Entrepreneurs sell each new firm at its expected value before the draw and remain in the entrepreneur population. New firms enter only the fringe; a fringe firm becomes a superstar only through a successful breakthrough innovation. Fringe firms die exogenously at Poisson rate δ; firms that are not producing stay in the pool until they die and resume production when the industry's state makes it profitable. A superstar exits, with zero value, when a rival's innovation would put it more than $\bar{n}$ steps behind the leader, or when it is displaced by an entrant (Section 4.2).

Modeling entry into and exit from the competitive fringe explicitly permits a realistic mapping between fringe-firm entry in the model and new business creation in the data. This contrasts with the papers where entrants either immediately become industry leaders or are otherwise large enough to interact strategically with incumbents. Instead, our model features the following firm life cycle: a large pool of fringe firms in each industry, of which only part produces at any time, and an even smaller subset that eventually breaks through into the superstar set.

4.2 Innovation

Innovation occurs through two distinct technologies: incremental innovation by incumbent superstars on the quality ladder, and breakthrough innovation by active fringe firms to become a superstar.

Superstar incremental innovation. Each superstar i in industry j chooses an R&D intensity $z_{ijt} \geq 0$ that generates a Poisson arrival rate z_{ijt} of a one-step quality improvement, at cost

$$\text{cost}_i^{\text{R\&D}} \;=\; \chi z_{ijt}^{\phi} Y_t, \qquad \phi > 1. \tag{8}$$

Upon success, the firm's productivity rises by a factor $(1+\lambda)$. If the leader, or a firm tied with it, innovates, the frontier moves up one step and every other superstar's gap rises by one; a superstar

whose gap would exceed $\bar{n}$ exits.

Fringe breakthrough innovation. Each active fringe firm k chooses an arrival rate $x_{ckjt} \geq 0$ of a breakthrough at cost

$$\text{cost}_k^{\text{breakthrough}} \;=\; \chi_x \, x_{ckjt}^{\phi_x} \, Y_t, \qquad \phi_x > 1. \tag{9}$$

Fringe firms that are not producing do not innovate. All active fringe firms in an industry choose the same rate (Section 4.5), so the industry-level rate of breakthroughs is $X_{jt} = m_{jt}^{\text{a}} \, x_{cjt}$.

Where a new superstar lands. Let the bar B be the gap of the $\bar{N}$-th superstar when the industry is full ($N = \bar{N}$), and the boundary $\bar{n}$ when a seat is free ($N < \bar{N}$). A successful entrant starts one step below the bar and climbs $k \geq 1$ steps with probability $(1-q)\,q^{k-1}$, landing at gap $L = B + 1 - k$. A landing $L < 0$ is a leapfrog: the entrant becomes the new leader, $|L|$ steps ahead of the former leader; every incumbent's gap rises by $|L|$, incumbents pushed beyond $\bar{n}$ exit, and the frontier advances by $|L|$ steps. Landings more than $\bar{\ell}$ steps ahead of the former leader are assigned to $L = -\bar{\ell}$ ($\bar{\ell} = 3$). The landing probabilities are therefore

$$p_L \;=\; (1-q)\,q^{B-L} \quad \text{for } -\bar{\ell} < L \leq B, \qquad p_{-\bar{\ell}} \;=\; q^{B+\bar{\ell}}. \tag{10}$$

When the industry is full, every landing ties or passes the $\bar{N}$-th superstar, and the incumbent at the bottom gap is displaced and exits (one of them at random when several are tied there). When a seat is free, the entrant takes it and no incumbent is displaced. An entrant tied with an incumbent ranks behind it.

4.3 Lobbying

Firms can lobby on three fronts $C \in \{T, S, R\}$: their own profit tax (T), their own procurement contracts (S), and the industry-wide regulatory fixed cost (R). A superstar on front C chooses an intensity $b_{Ci} \geq 0$ at cost $\kappa_C \, b_{Ci}^{\phi_C} \, Y_t$, with $\phi_C > 1$. Lobbying costs are not deductible from the profit-tax base.[2]

Profit tax (T). Lobbying on the tax front reduces the firm's own effective tax rate from the statutory τ_0:

$$\tau_{ijt} \;=\; \frac{\tau_0}{1 + a_{Tijt} \, b_{Tijt}}, \tag{11}$$

where $a_{Tijt} \in \{0, 1\}$ indicates that the firm lobbies on the front. The rate applies to the firm's deductible base, defined in Section 4.5, with full loss offset: a negative base earns a refund.

[2] Internal Revenue Code §162(e) disallows the deduction of lobbying expenditures for federal income tax purposes.

Procurement (S). Superstar i holds government contracts worth

$$g_{ijt} \;=\; \left(g_0 + a_{Sijt}\, b_{Sijt}\right)(1+\lambda)^{-\kappa_g n_{ijt}}\; Y_t, \tag{12}$$

where g_0 is the baseline contract of every superstar, b_{Sijt} is the increment won by lobbying, and κ_g governs how contracts fall with the firm's distance to the leader. Contracts are cost-plus: the firm delivers goods to the government at cost, $(1-\theta_g)\, g_{ijt}$, and keeps the fee $\theta_g\, g_{ijt}$ as taxable profit. Fringe firms hold no contracts.

Regulatory fixed cost (R): the contest. The industry-wide regulatory cost is determined by an influence contest:

$$R_{jt} \;=\; R_0\,\frac{1+\beta^p\, B^{pro}_{jt}}{1+\beta^a\, B^{anti}_{jt}}, \qquad B^s_{jt} \;=\; \sum_{i\in s} b^\theta_{Rijt} \;+\; b^\theta_{Ajt}\,\mathbf{1}\{\text{association on side } s\}, \qquad \theta\in(0,1), \tag{13}$$

where B^{pro}_{jt} and B^{anti}_{jt} collect the influence of the superstars and of the fringe's association on each side, and b_{Ajt} is the association's intensity. We normalize $\beta^p = 1$: the scale of influence is in the cost κ_R. The parameter β^a measures how effective lobbying against the cost is relative to lobbying for it. Without lobbying, $R_{jt} = R_0$, where $R_0(\kappa_c \bar{N} + \bar{m}^{\mathrm{a}}) = c$, so that compliance costs are a share c of output, with $\bar{m}^{\mathrm{a}}$ the observed number of active fringe firms per industry. When the opposing side dominates, R_{jt} falls below R_0. Each participant chooses its side by the sign of its stake in R (Section 4.5). Superstars pay $\kappa_R\, b^{\phi_R}_{Ri}\, Y_t$ for intensity b_{Ri}.

The contest and legislation. The influence function (13) summarizes a legislative process. The regulatory fixed cost of an industry is the stock of rules in force; proposals that raise it and proposals that lower it arrive over time, rules lapse at a constant rate, and each proposal passes with a probability that rises with the lobbying force of the firms that push it and falls with the force of those that block it, as in Kang (2016). A firm's force is $\beta^p b^\theta_{Ri}$ on the pro side and $\beta^a b^\theta_{Ri}$ on the anti side. Appendix E shows that in the steady state of this process the stock of rules relative to its level without lobbying equals $1 + \beta^p B^{pro}_{jt} - \beta^a B^{anti}_{jt}$ to first order in the forces, which is the first-order expansion of (13). This equivalence is what allows us to identify the contest from the enactment of bills (Section 5.5). The multiplicative form keeps R_{jt} positive and lets either side move it: a dominant pro side raises the cost above R_0, and a dominant anti side lowers it below.

Participation in lobbying. To lobby on front C, a superstar pays a flow fixed cost $F_{Ci}\, Y_t$, with

$$\ln F_{Ci} \;\sim\; N(\mu_C, \sigma^2_C), \qquad \text{i.i.d. across firms and fronts.} \tag{14}$$

A new draw arrives at Poisson rate λ_C, and only then can the firm change its status $a_{Ci} \in \{0,1\}$. A firm that joins pays its draw until the next one arrives or until it exits. Because the draws are independent across firms and fronts, no two firms revise at the same instant. A firm on front C that

is also on n_{-C} of its other fronts pays $F_{Ci}\,(1-\iota)^{n_{-C}}$, with $\iota \in [0, 1/3)$: lobbying on several fronts shares organizational costs (the bound keeps the marginal cost of a third front positive; $\iota = 0$ in the baseline).

Fringe trade association. Fringe firms are atomistic and cannot lobby individually: each firm has measure zero and therefore zero influence. They can lobby on the regulatory cost through a trade association of the industry. The association organizes with the same timing as a superstar's participation: at Poisson rate λ_A a new organizing cost $F_A\,Y_t$ is drawn, $\ln F_A \sim N(\mu_A, \sigma_A^2)$; if the association organizes, it pays the cost until the next draw. Its status is $a_{Ajt} \in \{0, 1\}$, and the organizing cost is shared equally by all firms in the industry's fringe pool. When organized, the association chooses its intensity b_{Ajt} every instant, at cost

$$C_A\big(n_{jt}, b_{Ajt}\big) \;=\; \kappa_A\, n_{jt}^{\phi_c}\, b_{Ajt}^{\phi_R}\, Y_t, \tag{15}$$

financed by the dues of its n_{jt} members. With $\phi_c > 1$, coordinating a larger membership requires disproportionately more communication, monitoring of contributions, and resolution of internal disagreements, so the association becomes less efficient as it grows (Olson, 1965). Dues are proportional to each member's operating surplus, $d_{jt}(\varepsilon) \propto \varepsilon_{jt}^{*} - \varepsilon$, so the marginal active firm pays nothing and every active firm is a member: $n_{jt} = m_{jt}^{\mathrm{a}}$. The dues of all members together equal C_A.

4.4 Government and households

Government. The government buys a fixed amount of final goods through procurement contracts: total contract outlays are $\bar{G}\,Y_t$. Superstars' contracts $D_t = \int_0^1 \sum_i g_{ijt}\,dj$ come out of this budget, and the rest, $\bar{G}Y_t - D_t$, goes to contractors outside the model's superstars, whose fee is a transfer to households. Contracts are bought at cost, so government purchases are $G_t = (1-\theta_g)\,\bar{G}\,Y_t$. Corporate tax revenue at the statutory rate is rebated lump-sum. A labor income tax τ_w finances the purchases and the revenue lost to tax lobbying:

$$\tau_w\, w_t\, L_t \;=\; \bar{G}\,Y_t \;+\; \int_0^1 \sum_{i=1}^{N_{jt}} \big(\tau_0 - \tau_{ijt}\big)\, \mathcal{B}_{ijt}\, Y_t\; dj, \tag{16}$$

where $\mathcal{B}_{ijt}$ is the deductible base of superstar i (in units of Y_t), defined below. Compliance costs are a resource cost, not government revenue.

Households. The representative household chooses private consumption and labor supply, taking government purchases as given:

$$\max_{\{C_t^p,\, L_t\}} \int_0^\infty e^{-\rho t} \left[\ln\big(C_t^p + G_t\big) - \xi\, \frac{L_t^{\,1+1/\epsilon_L}}{1+1/\epsilon_L} \right] dt \quad \text{s.t.} \quad \dot{A}_t = r_t A_t + (1-\tau_w)\, w_t L_t + T_t - C_t^p,$$

where A_t is wealth, r_t the return on assets, w_t the wage, and T_t lump-sum transfers. With Λ_t the marginal value of wealth, the first-order conditions are $e^{-\rho t}/(C_t^p + G_t) = \Lambda_t$, $\dot{\Lambda}_t = -r_t \Lambda_t$ and $e^{-\rho t}\, \xi L_t^{1/\epsilon_L} = \Lambda_t(1-\tau_w)w_t$, which give the Euler equation for total consumption and labor supply:

$$\frac{\dot{C}_t}{C_t} \;=\; r_t - \rho, \qquad \xi\, L_t^{1/\epsilon_L} \;=\; \frac{(1-\tau_w)\, w_t}{C_t}, \qquad C_t = C_t^p + G_t. \tag{17}$$

Private consumption and government purchases are both shares of output, so on the balanced growth path both grow at rate g and $r_t = \rho + g$. Every firm decision below is in units of Y_t and does not depend on the wage, labor supply or the labor tax; these determine only the level of output and welfare.

4.5 Decisions in the decentralized economy

Final good producer. A representative final good producer chooses the quantity of each variety in each industry competitively, which produces the inverse demand functions for superstars and the price of fringe output:

$$p_{ijt} = \frac{y_{ijt}^{-\frac{1}{\eta}} Y_t}{\sum_{k=1}^{N_{jt}} y_{kjt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{cjt}^{\frac{\eta-1}{\eta}}}, \qquad p_{cjt} = \frac{\tilde{y}_{cjt}^{-\frac{1}{\eta}} Y_t}{\sum_{k=1}^{N_{jt}} y_{kjt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{cjt}^{\frac{\eta-1}{\eta}}}. \tag{18}$$

Government purchases are purchases of the final good, so they do not change any industry's demand.

Product market. Superstars compete à la Cournot. Each superstar chooses y_{ijt} to maximize operating profit $\pi_{ijt} = p_{ijt}\, y_{ijt} - w_t\, l_{ijt}$, taking rivals' quantities and the fringe aggregate $\tilde{y}_{cjt}$ as given, subject to (4), (5) and (18). Fringe firms take p_{cjt} as given. Appendix A derives the best responses and profits. In terms of the superstar's share of industry sales, $\sigma_{ijt} = p_{ijt} y_{ijt}/Y_t$, its markup and operating profit are

$$M_{ijt} \;=\; \frac{\eta}{(\eta-1)(1-\sigma_{ijt})}, \qquad \pi_{ijt} \;=\; \sigma_{ijt}\Big(1 - \frac{\alpha}{M_{ijt}}\Big)\, Y_t, \tag{19}$$

and each active fringe firm earns

$$\pi_{cjt} \;=\; \frac{(1-\alpha)\, \sigma_{cjt}\, Y_t}{m_{jt}^{\mathrm{a}}}, \tag{20}$$

where σ_{cjt} is the fringe's share of industry sales. Fringe firms charge no markup, yet decreasing returns leave them positive operating profits. Given productivities, all sales shares depend only on the industry's gaps and the active fringe mass, and $\partial \pi_{ijt}/\partial m_{jt}^{\mathrm{a}} < 0$, $\partial \pi_{cjt}/\partial m_{jt}^{\mathrm{a}} < 0$.

Taxable income. The deductible base of a superstar is operating profit plus its contract fee, minus compliance and R&D (in units of Y_t):

$$\mathcal{B}_{ijt} \;=\; \frac{\pi_{ijt}}{Y_t} \;+\; \theta_g \left(g_0 + a_{Sijt}\, b_{Sijt}\right)(1+\lambda)^{-\kappa_g n_{ijt}} \;-\; \kappa_c R_{jt} \;-\; \chi z_{ijt}^{\phi}. \tag{21}$$

Fringe firms pay the statutory rate τ_0 on operating profit minus compliance, their fixed cost ε_k and their breakthrough R&D. Lobbying intensities, participation and organizing costs, and association dues are not deductible.

States. The *technology state* of an industry is $\Theta = (N, \tilde{\mathbf{n}})$, where N is the number of superstars and $\tilde{\mathbf{n}} = (\tilde{n}_1, \ldots, \tilde{n}_{N-1})$ is the sorted vector of follower gaps behind the leader, $0 \le \tilde{n}_1 \le \cdots \le \tilde{n}_{N-1} \le \bar{n}$. The *industry state* adds the lobbying statuses that determine the regulatory cost: $S = (\Theta, \mathbf{a}_R, a_A)$, where $\mathbf{a}_R = (a_{R1}, \ldots, a_{RN})$ records which superstars lobby on R and a_A whether the association is organized. These statuses are tracked exactly and move with the firms when ranks change; an entrant starts with every status off. A superstar's own state s_i adds its seat in S, its statuses on the tax and procurement fronts (a_{Ti}, a_{Si}), and the participation costs it has locked on each front. Rivals' tax and procurement statuses are not part of the state. Each firm holds beliefs $\pi(a_T, a_S \mid \Theta, \text{seat})$ about them, equal to their stationary frequencies among superstars in the same technology state and seat (Appendix C).[3] Values are normalized by output on the balanced growth path, $v \equiv V/Y_t$, and discounted at $r - g = \rho$; we drop time and industry subscripts where this creates no ambiguity.

Superstar problem. A superstar chooses its R&D intensity and its lobbying intensities on the fronts it is on every instant, and its participation on front C whenever its cost on that front is redrawn. Its Hamilton–Jacobi–Bellman equation is

$$
\begin{aligned}
\rho\, v(S, s_i) \;=\; \max_{z_i,\, b_{Ri},\, b_{Ti},\, b_{Si}} \Big\{ & (1-\tau_i)\,\mathcal{B}_i \;-\; \sum_{C \in \{R,T,S\}} a_{Ci}\Big[\kappa_C\, b_{Ci}^{\phi_C} + F_{Ci}\,(1-\iota)^{n_{-C}}\Big] \\
& + z_i\left[v(S^{+i}, s_i^{+i}) - v(S, s_i)\right] \;+\; \sum_{k \neq i} \bar{z}_k(S)\left[v(S^{+k}, s_i^{+k}) - v(S, s_i)\right] \\
& + X(S) \sum_{L} p_L(S)\left[v(S^{e,L}, s_i^{e,L}) - v(S, s_i)\right] \\
& + \sum_{k \neq i} \lambda_R\, \bar{p}_k^R(S)\left[v(S^{\neg k}, s_i) - v(S, s_i)\right] \\
& + \lambda_A\, p^A(S)\left[v(S^{\neg A}, s_i) - v(S, s_i)\right] \\
& + \sum_{C \in \{R,T,S\}} \lambda_C \Big(\mathbb{E}_{F'} \max\left\{ v^{C,\text{on}}(F'),\; v^{C,\text{off}} \right\} - v(S, s_i) \Big) \Big\},
\end{aligned}
\tag{22}
$$

subject to (13). The first line is the flow: after-tax income, and lobbying intensities and locked participation costs on the fronts the firm is on. The second line contains the firm's own innovation, after which the state is (S^{+i}, s_i^{+i}), and rivals' innovation at their belief-weighted rates $\bar{z}_k(S)$. The third line is entry: a fringe breakthrough arrives at rate $X(S)$ and lands at L with probability (10). The fourth and fifth lines contain rivals' redraws on R, which flip rival k's status with belief-weighted probability $\bar{p}_k^R(S)$, and the association's redraws, which flip its status with probability $p^A(S)$. The

[3] These are analogy-based expectations in the sense of Jehiel (2005), with classes defined by the technology state and the seat.

last line contains the firm's own redraws: $v^{C,\text{on}}(F')$ is the value after joining front C with locked cost F', and $v^{C,\text{off}}$ the value after staying out (on R the status is part of the industry state S; on T and S it is part of the firm's own state). A firm that exits, by falling more than $\bar{n}$ steps behind or by displacement, has value zero. The first-order conditions for innovation and procurement lobbying are

$$(1-\tau_i)\,\chi\phi\,z_i^{\phi-1} \;=\; v(S^{+i},s_i^{+i})-v(S,s_i), \qquad (1-\tau_i)\,\theta_g\,(1+\lambda)^{-\kappa_g n_i} \;=\; \kappa_S\,\phi_S\,b_{Si}^{\phi_S-1}. \tag{23}$$

The tax intensity b_{Ti} is chosen jointly with z_i and b_{Si} to maximize the right-hand side of (22); at an interior optimum,

$$\frac{\tau_0\,\mathcal{B}_i}{(1+b_{Ti})^2} \;=\; \kappa_T\,\phi_T\,b_{Ti}^{\phi_T-1}, \tag{24}$$

and because $\mathcal{B}_i$ depends on b_{Ti} through z_i and b_{Si}, we take the global maximum. The value is affine in the locked participation costs: $v = v^0 - \sum_C a_{Ci}F_{Ci}D_C$, where $D_C(S,s_i)$ is the expected discounted time over which the current draw on front C is paid. At a redraw, the firm joins front C if and only if its new draw is below a cutoff:

$$a'_{Ci}=1 \iff F' \;\le\; F_C^*(S,s_i) \;=\; \frac{v^0_{C,\text{on}}-v^0_{C,\text{off}}-\sum_{C'\neq C}a_{C'i}\,F_{C'i}\left[D_{C',\text{on}}-D_{C',\text{off}}\right]}{D_{C,\text{on}}}, \tag{25}$$

where the subscripts on and off denote the firm's state after joining and after staying out. The cross terms are zero when $\iota = 0$. Appendix B gives the durations and the value equations.

The stake in regulation. A superstar on the R front chooses b_{Ri} to satisfy

$$\Delta_i(S,s_i)\,\frac{\partial R(S)}{\partial b_{Ri}} \;=\; \kappa_R\,\phi_R\,b_{Ri}^{\phi_R-1}, \tag{26}$$

where Δ_i is the marginal value of a higher R to the firm:

$$\Delta_i \;=\; \underbrace{-(1-\tau_i)\,\kappa_c}_{\text{(i) own compliance}} \;+\; \underbrace{(1-\tau_i)\,\frac{\partial(\pi_i/Y)}{\partial m^{\text{a}}}\,\frac{dm^{\text{a}}}{dR}}_{\text{(ii) fringe exit: Cournot gain}} \;+\; \underbrace{\frac{dX}{dR}\sum_L p_L\left[v(S^{e,L},s_i^{e,L})-v(S,s_i)\right]}_{\text{(iii) fewer entrants: displacement risk}}. \tag{27}$$

Equation (27) is the central object of the paper: it is the shadow value of regulation to an incumbent, and its sign determines which side of the contest the firm joins. Channel (i) is the firm's own compliance cost, after tax because compliance is deductible; it is negative. Channel (ii) is the static Cournot gain as marginal fringe firms stop producing; it is positive, since $dm^{\text{a}}/dR < 0$ and $\partial\pi_i/\partial m^{\text{a}} < 0$. Channel (iii) is the lower rate of fringe breakthroughs, which would push the firm down the ranks or displace it; it is positive, since $dX/dR < 0$ and an entry lowers the incumbent's value. In a Markov-perfect equilibrium each agent takes rivals' policies, the fringe's breakthrough intensity and continuation values as given. The only fringe firm whose status responds to R is the

marginal one at ε^*, so

$$\frac{dm^{\mathrm{a}}}{dR} \;=\; m\, f(\varepsilon^*)\,\frac{d\varepsilon^*}{dR}, \qquad \frac{d\varepsilon^*}{dR} \;=\; \frac{-1}{1 - m\, f(\varepsilon^*)\,\dfrac{\partial(\pi_c/Y)}{\partial m^{\mathrm{a}}}}, \qquad \frac{dX}{dR} \;=\; x_c\,\frac{dm^{\mathrm{a}}}{dR}. \tag{28}$$

The denominator exceeds one, since $\partial\pi_c/\partial m^{\mathrm{a}} < 0$: as R removes marginal fringe firms, the survivors earn higher profits, which partially offsets the direct compliance burden. The contest uses the stake averaged over the firm's own tax and procurement statuses with the belief weights, so a superstar's barrier intensity does not depend on its own tax or procurement status (Appendix C). The firm lobbies for a higher cost if $\Delta_i > 0$ and against it if $\Delta_i < 0$:

$$\mathcal{P}(S) = \{i : a_{Ri} = 1,\ \Delta_i > 0\}, \qquad \mathcal{A}(S) = \{i : a_{Ri} = 1,\ \Delta_i < 0\}. \tag{29}$$

Given the influence function (13), the marginal influence of b_{Ri} is

$$\frac{\partial R(S)}{\partial b_{Ri}} \;=\; \begin{cases} \dfrac{R_0\,\beta^p\,\theta\,b_{Ri}^{\theta-1}}{1+\beta^a B^{\mathrm{anti}}} & \text{if } i \in \mathcal{P}(S), \\[2ex] -\dfrac{R\,\beta^a\,\theta\,b_{Ri}^{\theta-1}}{1+\beta^a B^{\mathrm{anti}}} & \text{if } i \in \mathcal{A}(S), \end{cases} \tag{30}$$

so that (26) gives $b_{Ri} = \big[\Delta_i R_0 \beta^p \theta/(\kappa_R \phi_R (1+\beta^a B^{\mathrm{anti}}))\big]^{1/(\phi_R-\theta)}$ on the pro side and $b_{Ri} = \big[|\Delta_i| R \beta^a \theta/(\kappa_R \phi_R (1+\beta^a B^{\mathrm{anti}}))\big]^{1/(\phi_R-\theta)}$ on the anti side, which requires $\phi_R > \theta$. On the anti side the right-hand side depends on the firm's own intensity through R and B^{anti}, so the intensities at each state solve a fixed point.

Fringe problem. Within an industry, all fringe firms share the same productivity and earn the same operating profit $\pi_c(S)$; they differ only in ε. The HJB equation of a fringe firm with cost draw ε is

$$\begin{aligned}(\rho+\delta)\,v^e(S,\varepsilon) \;=\; \max_{a,\,x}\, a\Big\{(1-\tau_0)\Big[\frac{\pi_c(S)}{Y} - R(S) - \varepsilon - \chi_x x^{\phi_x}\Big] \;-\; a_A\, d(S,\varepsilon) \\ + x\, v^{\mathrm{entry}}(S)\Big\} \;-\; a_A\,\hat{f}_A(S) \;+\; \sum_{S'\neq S} p(S,S')\,\big[v^e(S',\varepsilon) - v^e(S,\varepsilon)\big],\end{aligned} \tag{31}$$

where $a \in \{0,1\}$ indicates that the firm produces, $x \geq 0$ is its breakthrough intensity, $d(S,\varepsilon)$ are its dues, $\hat{f}_A(S)$ is its share of the association's locked organizing cost, and $p(S,S')$ are the industry's transition rates, induced by superstars' innovation, breakthroughs, and changes in lobbying statuses. A successful breakthrough makes the firm a new superstar with all lobbying statuses off; its expected value over the landing (10) is

$$v^{\mathrm{entry}}(S) \;=\; \sum_L p_L(S)\; v\big(S^{e,L}, \text{entrant's seat}\big). \tag{32}$$

The first-order condition gives the same breakthrough intensity for every active firm in the industry,

$$(1-\tau_0)\,\chi_x\,\phi_x\,x_c^{\phi_x-1} \;=\; v^{\text{entry}}(S). \tag{33}$$

Because switching is costless and ε is permanent, the decision to produce affects only the current flow. A fringe firm produces if and only if $\varepsilon \le \varepsilon^*(S)$, where

$$\varepsilon^*(S) \;=\; \frac{\pi_c(S)}{Y} - R(S) + \frac{\Omega(S)}{1-\tau_0}, \qquad \Omega(S) \;=\; \max_{x\ge 0}\left\{x\,v^{\text{entry}}(S) - (1-\tau_0)\,\chi_x x^{\phi_x}\right\}. \tag{34}$$

Dues do not enter the cutoff, since the marginal firm pays none. Because π_c depends on $m^{\text{a}} = mF(\varepsilon^*)$, (34) is a fixed point at each state; it has a unique solution, since π_c falls with m^{a}.

Let $\bar{v}^e(S) \equiv \mathbb{E}_\varepsilon[v^e(S,\varepsilon)]$ be the expected value of a firm in the industry's fringe pool. Integrating (31) over ε,

$$(\rho+\delta)\,\bar{v}^e(S) \;=\; (1-\tau_0)\,\mathbb{E}\big[(\varepsilon^*(S)-\varepsilon)^+\big] \;-\; \frac{a_A\,C_A(S)}{m\,Y} \;-\; a_A\,\hat{f}_A(S) \;+\; \sum_{S'\neq S} p(S,S')\,\big[\bar{v}^e(S')-\bar{v}^e(S)\big], \tag{35}$$

where the second term is total dues per pool firm. Entrepreneurs sell new firms at $\mathbb{E}_\mu[\bar{v}^e]$, the pool value averaged over the stationary distribution μ of industries. With $e = \delta m$ on the balanced growth path, the entrepreneurs' first-order condition gives the free-entry condition

$$2\psi\delta\,m \;=\; \sum_S \mu(S)\,\bar{v}^e(S). \tag{36}$$

The entry cost ψ is calibrated so that, without lobbying, the number of active fringe firms per industry equals its observed value $\bar{m}^{\text{a}}$; it is held fixed in every counterfactual, in which m solves (36).

Association problem. At a redraw of its organizing cost, the association organizes if and only if

$$F_A \;\le\; F_A^*(S) \;=\; (\rho+\lambda_A)\,m\,\big[\bar{v}^{e,\text{on}}(S) - \bar{v}^{e,\text{off}}(S)\big], \tag{37}$$

where $\bar{v}^{e,\text{on}}$ and $\bar{v}^{e,\text{off}}$ are the pool values with the association organized and not, the former gross of the current organizing cost. When organized, the association chooses b_A to maximize its members' total gain from the regulatory cost net of the lobbying cost, taking rivals' intensities as given:

$$\max_{b_A\ge 0}\;\; n(S)\,\Delta_A(S)\,R(b_A,\mathbf{b}_{-A}) \;-\; \kappa_A\,n(S,R)^{\phi_c}\,b_A^{\phi_R}, \tag{38}$$

where membership responds to R through $n = m^{\text{a}}$. The first-order condition is

$$n\,\Delta_A\,\frac{\partial R}{\partial b_A} \;=\; \kappa_A\,\phi_R\,n^{\phi_c}\,b_A^{\phi_R-1} \;+\; \phi_c\,\bar{d}\,\frac{\partial n}{\partial b_A}, \qquad \bar{d} = \kappa_A\,n^{\phi_c-1}\,b_A^{\phi_R}, \qquad \frac{\partial n}{\partial b_A} = \frac{dm^{\text{a}}}{dR}\,\frac{\partial R}{\partial b_A}, \tag{39}$$

where $\bar{d}$ are dues per member and $\partial R/\partial b_A$ takes the form (30) for the association's side. The last term is the Olson friction: a lobbying effort that changes membership changes the cost of coordinating it. The marginal value of a higher R to a member is

$$\Delta_A \;=\; \underbrace{-(1-\tau_0)}_{\text{own compliance}} \;+\; \underbrace{(1-\tau_0)\,\frac{\partial(\pi_c/Y)}{\partial m^{\mathrm{a}}}\,\frac{dm^{\mathrm{a}}}{dR}}_{\text{fringe exit: survivors' profit}} \;+\; \underbrace{\frac{m}{m^{\mathrm{a}}}\,\frac{dX}{dR}\sum_{L} p_L\left[\bar{v}^{e}(S^{e,L})-\bar{v}^{e}(S)\right]}_{\text{breakthroughs: the pool's value}}. \tag{40}$$

By (34), the first two terms sum to $(1-\tau_0)\,d\varepsilon^*/dR < 0$: the association can favor a higher cost only through the last term, which captures how a breakthrough changes the value of the industry's fringe position. The association lobbies on the side given by the sign of Δ_A.

Within-state political conflict. The contest at state S exhibits a within-industry split whenever some superstars have $\Delta_i > 0$ while the association has $\Delta_A < 0$. The two coalitions push in opposite directions through (13); the equilibrium $R(S)$ is pinned down by the joint fixed point of the superstar conditions (26) and the association condition (39).

4.6 Equilibrium

Definition 1 (*Balanced-Growth-Path Equilibrium*)**.** *A balanced-growth-path (BGP) equilibrium consists of*

- superstar values $v(S, s_i)$ and policies $\{z_i, b_{Ri}, b_{Ti}, b_{Si}\}$ with participation cutoffs $\{F_C^*\}$; fringe values $v^e(S,\varepsilon)$, cutoffs $\varepsilon^*(S)$ and breakthrough intensities $x_c(S)$; the association's cutoff $F_A^*(S)$ and intensity $b_A(S)$;
- the regulatory cost $R(S)$ at every industry state;
- beliefs $\pi(a_T, a_S \mid \Theta, \text{seat})$;
- a fringe pool m and active masses $m^{\mathrm{a}}(S) = mF(\varepsilon^*(S))$;
- a stationary distribution μ over industry states and superstars' own statuses;
- a wage ratio $\omega = w/Y$, labor supply L, a labor tax τ_w, and a growth rate g;

such that:

1. **Product market.** Given $m^{\mathrm{a}}(S)$, quantities and prices satisfy the superstars' Cournot best responses and the fringe's price-taking conditions.
2. **Superstars.** Values and policies solve (22) given $R(\cdot)$, the beliefs and rivals' strategies; participation follows (25).
3. **Fringe.** $\varepsilon^*(S)$ satisfies (34), $x_c(S)$ satisfies (33), and the association's decisions satisfy (37) and (39).

4. **Contest.** At every state, intensities satisfy (26) for every superstar on the R front, with sides (29), and $R(S)$ is given by (13).

5. **Free entry.** m satisfies (36).

6. **Consistency.** μ is the invariant distribution of the process induced by the policies, and the beliefs equal its stationary frequencies.

7. **Labor market and government.** Labor demand equals labor supply (17), and the government budget (16) balances.

Growth. The frontier of an industry advances when its leader, or a firm tied with it, innovates, and when an entrant leapfrogs the leader. The aggregate growth rate is

$$g \;=\; \ln(1+\lambda)\,\sum_{S}\mu(S)\Big[\sum_{i:\,n_i=0} z_i(S) \;+\; X(S)\sum_{L<0} p_L(S)\,|L|\Big]. \tag{41}$$

Labor market and output. Within each industry, producers' labor demands follow from the static stage; each is proportional to Y/w. Labor market clearing requires

$$\int_0^1 \left(\sum_{i=1}^{N_{jt}} l_{ijt} + \int_0^{m_t} l_{ckjt}\,dk\right) dj \;=\; L_t, \tag{42}$$

with L_t given by the household's labor supply (17). Output then follows from (2).

Resource constraint. Final-good output is allocated between private consumption, government purchases, R&D, entry, fixed costs, compliance and lobbying. Dividing by Y_t and aggregating across industries with μ, the BGP resource constraint reads

$$1 \;=\; \frac{C_t^p}{Y_t} \;+\; (1-\theta_g)\,\bar{G} \;+\; \mathcal{R}_{\text{R\&D}} \;+\; \psi\,(\delta m)^2 \;+\; \mathcal{R}_{\text{fixed}} \;+\; \mathcal{R}_{\text{comp}} \;+\; \mathcal{L}, \tag{43}$$

where

$$\begin{aligned}
\mathcal{R}_{\text{R\&D}} \;&=\; \mathbb{E}_\mu\Big[\sum_i \chi\, z_i^{\phi}\Big] \;+\; \sum_S \mu(S)\, m^{\text{a}}(S)\,\chi_x\, x_c(S)^{\phi_x},\\
\mathcal{R}_{\text{fixed}} \;&=\; m\sum_S \mu(S)\,\mathbb{E}\big[\varepsilon\,\mathbf{1}\{\varepsilon \le \varepsilon^*(S)\}\big],\\
\mathcal{R}_{\text{comp}} \;&=\; \sum_S \mu(S)\,\big[\kappa_c N(S) + m^{\text{a}}(S)\big]\,R(S),\\
\mathcal{L} \;&=\; \mathbb{E}_\mu\Big[\sum_i\sum_C a_{Ci}\big(\kappa_C\, b_{Ci}^{\phi_C} + F_{Ci}\,(1-\iota)^{n-C}\big)\Big] \;+\; \sum_S \mu(S)\,a_A\Big[\frac{C_A(S)}{Y} + F_A\Big].
\end{aligned}$$

The expectations $\mathbb{E}_\mu$ run over industry states and over superstars' own statuses and locked costs. Compliance is a pure resource cost: paying it generates no output, consistent with the interpretation

of regulation as administrative friction. Tax revenue, contract fees and association dues are transfers and do not appear in (43).

4.7 Welfare and Counterfactual Analysis

On the balanced growth path, total consumption grows at rate g, and welfare is

$$W \;=\; \frac{1}{\rho}\Big[\ln Y_0 + \ln\Big(\frac{C_0^p + G_0}{Y_0}\Big)\Big] \;+\; \frac{g}{\rho^2} \;-\; \frac{1}{\rho}\,\xi\,\frac{L^{1+1/\epsilon_L}}{1+1/\epsilon_L}. \tag{44}$$

We measure the welfare cost of lobbying by the consumption-equivalent variation relative to a reference economy without lobbying: the same economy with every participation cost set to infinity, so that $R = R_0$ and contracts equal their baseline, at the same tax rate τ_0, procurement budget and entry cost ψ, with m given by free entry. The variation solves

$$\ln(1 + \mathrm{CEV}) \;=\; \rho\,\big(W^{\mathrm{ref}} - W\big), \tag{45}$$

so that $\mathrm{CEV} > 0$ means that lobbying lowers welfare. A counterfactual ban on a front sets that front's participation costs to infinity.

Decomposing the welfare cost. Every producer's technology is homogeneous of degree α in labor, so output can be written exactly as

$$\ln Y_0 \;=\; \alpha \ln L + \ln A^* + \ln E, \qquad \ln A^* \;=\; \sum_S \mu(S)\,\frac{1}{e}\ln \sum_k A_k(S)^e, \qquad e = \frac{s}{1-\alpha s}, \quad s = \frac{\eta-1}{\eta}. \tag{46}$$

Here A^* is the productivity that an efficient allocation of the same labor would achieve given each industry's state and its operating fringe; $A_k = q_k$ for a superstar and $A_c = (m^{\mathrm{a}})^{1-\alpha} q_c$ for the fringe taken as one producer, with qualities relative to the industry's frontier. The term $\ln E \le 0$ is the allocative efficiency of the equilibrium allocation of labor across producers within and across industries, which markups distort. Substituting into (44), $100\ln(1+\mathrm{CEV})$ splits exactly into labor input, the efficient productivity of the industry states (the number of superstars and their gaps) and of the operating fringe, allocative efficiency, the resources absorbed by compliance, fixed costs, entry, R&D and lobbying (the change in C/Y), growth, and the disutility of labor. Appendix F gives the details.

5 Estimation

We estimate the model by the simulated method of moments. The parameters fall into three blocks, each disciplined by a different kind of data: the technology block by firm dynamics and industry structure; the tax and procurement block by the outcomes that lobbying buys, which the data reveal directly; and the regulation block, whose outcome is not observed, by participation,

sides, the trade associations, and the enactment of bills. All blocks are estimated jointly, because lobbying changes the industry structure that the technology moments describe and the industry structure determines the stakes that drive lobbying.

5.1 Strategy

Objective. Let $\hat{m}_k$ be the data value of moment k and $m_k(\vartheta)$ its model counterpart at parameters ϑ. The estimator minimizes the sum of squared deviations $\sum_k d_k(\vartheta)^2$ with equal weights, where $d_k = \ln(m_k/\hat{m}_k)$ for moments that are positive by construction (shares, rates, ratios, and levels) and $d_k = (m_k - \hat{m}_k)/\max(|\hat{m}_k|, \mathrm{se}_k)$ for moments that can take either sign (regression coefficients, elasticities, and differences). Deviations are capped at a hundredfold miss in either direction, so no single moment can dominate. The growth rate enters as a constraint: we keep only parameter values whose growth rate is within 8 percent of the data's (about one standard error). There are 106 targeted moments: 36 in the technology block, 62 lobbying moments, and 8 moments of the bill simulation; two further moments enter as upper bounds.

Model moments. Every model moment is computed with the definition of its data counterpart. Moments of the industry structure are computed from the stationary distribution over industry states. Moments that the data measure on a panel of firms, such as participation by rank, the persistence of lobbying, the participation of firms by decile of their stake, and the regressions of tax rates and contracts on past lobbying, are computed on a simulated panel of superstars of the same length, with the same regressions and fixed effects. Bill moments are computed by simulating every real proposal (Section 5.5).

Search. The model is costly to solve: at the estimate, a certified equilibrium takes about twenty minutes on one machine, and the objective is not smooth in all parameters. We therefore estimate the model by a global search. We draw 8,192 points from a scrambled Sobol sequence over a 37-dimensional box: the 11 technology parameters within five standard errors of an estimate of the technology block alone, 13 parameters of the regulation block, and 13 of the tax and procurement block, the latter drawn in the units of the outcomes they produce (the leader's tax cut, its share of contracts due to lobbying, its participation) and mapped to the structural cost parameters at each point. Each point is first screened on the fit of its technology block; 3,051 of the points that pass the screen are solved to a certified equilibrium (Appendix D), and 489 of them satisfy the growth constraint. We take as the estimate the best-fitting of these points, with one qualification discussed in Section 6.5: the two best points are within the simulation noise of the objective, and we select the one whose technology block fits far better. Section 6.5 also reports how the results vary across the best-fitting points; a local refinement around the estimate and standard errors are in progress.

Table 5. Parameters Set Outside the Model

Parameter		Value	Source
Discount rate	ρ	0.04	Cavenaile et al. (2026)
Returns to scale in labor	α	0.90	Cavenaile et al. (2026)
Frisch elasticity of labor supply	ϵ_L	0.5	standard
Maximum number of superstars	$\bar{N}$	4	superstar definition
Maximum gap behind the leader	$\bar{n}$	5	10 percent of the leader's sales
Active fringe firms per industry, no lobbying	$\bar{m}^{\mathrm{a}}$	5,013	BDS
Contractor's margin	θ_g	0.104	Compustat, federal contractors
Convexity of the barrier lobbying cost	ϕ_R	2	normalization
Effectiveness of pro-side lobbying	β^p	1	normalization
Largest leapfrog	$\bar{\ell}$	3	
Scope discount on participation costs	ι	0	
Capital share (labor-share mapping)		0.25	Barkai (2020)

Notes: The maximum gap is recomputed at every parameter point as the largest gap at which a superstar's sales are at least 10 percent of the leader's. The model has no physical capital; as in Cavenaile et al. (2026), its labor share is compared with the data after multiplication by one minus the capital share.

5.2 Parameters set outside the model

Table 5 lists the parameters we set outside the model. The discount rate is $\rho = 0.04$ and the degree of decreasing returns $\alpha = 0.9$, as in Cavenaile et al. (2026); the Frisch elasticity of labor supply is $\epsilon_L = 0.5$, and ξ normalizes labor supply to one in the economy without lobbying, taxes, and contracts. Each industry has at most $\bar{N} = 4$ superstars, as in the data's definition, and the maximum gap $\bar{n}$ is the largest gap at which a superstar still sells at least 10 percent of the leader's sales, the data's definition of a superstar; it is recomputed at every parameter point and equals $\bar{n} = 5$ at the estimate. The observed number of active fringe firms per industry, $\bar{m}^{\mathrm{a}} = 5{,}013$, is the average number of employer firms in a four-digit industry outside finance (Business Dynamics Statistics), and the entry cost ψ is set so that the economy without lobbying has this number. The contractor's margin $\theta_g = 0.104$ is the sales-weighted operating margin of firms whose federal contracts are at least 70 percent of their sales; it is between 10.3 and 10.5 percent for thresholds from 50 to 90 percent. The convexity of the barrier lobbying cost is $\phi_R = 2$, and we normalize $\beta^p = 1$, so that the scale of influence is carried by the price κ_R. Landings more than $\bar{\ell} = 3$ steps above the former leader are rare (Section 4.2), and the discount on the participation cost of a second front is set to $\iota = 0$.

5.3 Technology

The technology block has eleven parameters: the quality step λ and the elasticity of substitution η; the climb q of new superstars; the level χ and convexity ϕ of superstars' R&D cost; the level χ_x and convexity ϕ_x of the fringe's breakthrough cost; the fringe's relative productivity ζ; the level $\bar{\varepsilon}$ and dispersion σ of fringe fixed costs; and the fringe death rate δ. Cavenaile et al. (2026) discipline the quality ladder with patent data: an innovation–size regression and the leader's share of citations.

We rely mainly on firm dynamics measured in sales, complemented by the leaders' shares of R&D and of patents within pairs of top firms, for three reasons. Most firms in the regression have no patents, so the regression is driven by the extensive margin of patenting; the citation share implies dominant leaders that the sales data reject; and a fringe whose fixed costs and entry respond to regulation needs moments on fringe entry, exit, and losses, which patent data do not provide. The 36 technology moments are listed in Appendix G.

Identification. Every parameter affects every moment, but each moves some more than others. The distribution of sales gaps among the four largest firms (the leader's share of the top four, the ratios of the second and fourth firms to the leader, and the share of industries with fewer than four superstars) pins down the quality step λ and the elasticity η together: a larger step and a higher elasticity both widen sales gaps for given quality gaps. Growth, which rises with the step, and the superstars' markups, which fall with the elasticity, separate them. Where new superstars enter relative to the fourth firm, and their sales relative to the leader, pin down the climb q. The convexity of R&D costs ϕ is identified by growth, by how often a leader pulls away from a given follower within two years, and by the R&D and innovation shares of leaders. Its level χ is weakly identified, because R&D effort moves with $\chi^{-1/(\phi-1)} = \chi^{-0.19}$ at the estimate. The rate at which new firms join the top four pins down the fringe's breakthrough cost. Concentration among the four largest firms (Census CR4) and the share of fringe firm-years with losses pin down the fringe's productivity ζ. Business applications per firm, the share of applications that become employer firms, and firm exit rates pin down the level and dispersion of fringe fixed costs and the death rate δ: the first two measure how many potential firms find it worthwhile to operate, the third how fast the pool turns over.

Estimates and fit. Table 6 reports the estimates and Table 7 the fit of the main moments. The quality step is $\lambda = 0.158$, and the elasticity of substitution $\eta = 5.3$ implies a markup of 1.23 for a superstar with a negligible share and of 1.36 for the average leader. Fringe fixed costs are dispersed ($\sigma = 1.65$), so that at any time about a third of the fringe pool produces. The model matches the number of superstars per industry (3.63 against 3.60), the share of industries with fewer than four (0.26 against 0.25), the median ratio of the second firm to the leader (0.67 against 0.61), growth, superstars' markups (1.31 against 1.26), the exit rate of firms (0.090 against 0.079), and business applications per firm. The median technology moment is within 18 percent of its data value, and 21 of the 36 are within 25 percent. The main misses are three. The model's industries are less concentrated than in the Census (CR4 of 0.17 against 0.26), and the fourth firm is smaller relative to the leader (0.21 against 0.30). R&D spending is about a third of business-funded R&D in the data (0.7 against 2.0 percent of GDP). Fewer fringe firm-years have losses than in Compustat (0.11 against 0.28), and fixed costs among active fringe firms are less dispersed than in the data.

Table 6. Technology Parameters

Parameter		Estimate
Quality step	λ	0.158
Elasticity of substitution within the industry	η	5.30
Climb of new superstars	q	0.415
Superstar R&D cost: level	χ	6.18×10^3
Superstar R&D cost: convexity	ϕ	6.24
Fringe breakthrough cost: level	χ_x	44.0
Fringe breakthrough cost: convexity	ϕ_x	5.03
Fringe relative productivity	ζ	2.27
Fringe fixed cost: mean	$\bar{\varepsilon}$	1.02×10^{-4}
Fringe fixed cost: dispersion	σ	1.65
Fringe death rate	δ	0.089

Notes: Best point of the joint global search (Section 5.1). Costs are in units of aggregate output; the fringe's productivity is relative to the geometric mean of the superstar seats (6).

5.4 Tax and procurement

On the tax and procurement fronts the outcome of lobbying is in the data: we observe each firm's tax rate and its federal contracts before and after it lobbies. The block has thirteen parameters: the statutory tax rate τ_0; the baseline contract g_0 and its decline with the firm's gap κ_g; and, for each front, the lobbying cost κ_C and its convexity ϕ_C, and the mean, dispersion, and redraw rate of the participation cost $(\mu_C, \sigma_C, \lambda_C)$.

Identification. Table 8 summarizes the moments. What lobbying buys identifies the lobbying cost: within a firm, a year of tax lobbying lowers the next three years' book tax rate by 1.8 percentage points, about 7 percent of the rate, and a year of procurement lobbying raises the firm's prime contracts by 0.13 log points. The model computes the same regressions, with firm and year fixed effects, on its simulated panel. The level of the outcomes without lobbying identifies τ_0 (superstars' effective rate, 0.272), g_0 (contracts over sales, 0.022), and κ_g (the elasticity of contracts to sales across firms, 0.76). Who lobbies identifies the participation costs: participation by rank, and across deciles of the firm's stake (pretax income for the tax front, past contracts for procurement), rises from 2 to 72 percent on the tax front and from 2 to 59 percent on procurement. How long firms lobby identifies the redraw rates: 83 percent of tax lobbyers and 75 percent of procurement lobbyers still lobby a year later. How much they spend identifies the convexity of the lobbying cost: at an interior optimum, spending on a front equals the gain from lobbying divided by ϕ_C, and spending is 0.019 percent of the stake on the tax front and 0.17 percent on procurement. Spending far below the gain implies a very convex cost.

Estimates and fit. The statutory rate is $\tau_0 = 0.290$, at the upper end of its search range (0.272 to 0.290). Lobbying buys little at the estimate: a leader that lobbies on taxes lowers its rate by 0.42 percent of τ_0, and 5.2 percent of a lobbying leader's contracts are due to lobbying. Both lobbying

Table 7. Technology Block: Model and Data

Moment	Data	Model
Industry structure		
Superstars per industry	3.60	3.63
Industries with fewer than four superstars	0.250	0.264
Leader's share of top-four sales, median	0.463	0.480
#2 / #1 sales, median	0.609	0.671
#4 / #1 sales, median	0.295	0.210
CR4 (Census)	0.265	0.174
Growth, markups, R&D		
Growth rate	0.0166	0.0168
Markup, superstars (cost-weighted)	1.26	1.31
R&D / GDP	0.0197	0.0071
Labor share	0.596	0.647
Superstar dynamics		
Leader pulls away from a follower (two years)	0.068	0.042
New superstars per industry-year	0.211	0.127
New superstars entering at the #4's level	0.742	0.435
Fringe		
Fringe firm-years with losses	0.278	0.112
Firm exit rate (BDS)	0.079	0.090
Business applications per firm (BFS)	0.242	0.250
Applications that become employers (BFS)	0.274	0.359

Notes: Model values in the lobbying equilibrium at the estimate. Shown: 17 of the 36 technology moments, all weighted equally; Appendix Table 17 lists all of them.

costs are very convex ($\phi_T = 4.8$, $\phi_S = 24$). The model matches the level of contracts (0.024 against 0.022 of sales), spending relative to the stake on procurement (0.20 against 0.17 percent), and participation on the tax front by the second and third superstars. It misses in two ways. The gains from lobbying are too small: the tax rate falls by 0.05 percentage points after a year of lobbying, against 1.78 in the data, and contracts rise by 0.031 log points against 0.135. And participation is too concentrated at the top on the tax front and too low on procurement: the model's leader lobbies on taxes too often (0.38 against 0.27) and its fourth firm too rarely (0.05 against 0.12), and procurement participation is between a fifth and a half of the data's, falling short most at the bottom ranks (Figure 7). Because the estimated gains are small, the model's tax and procurement lobbying moves smaller transfers than the data suggest; Section 6.4 explains why transfers of this kind cost little in the model.

5.5 Regulation

The regulation block raises four difficulties, and we address each with its own source of data. First, the regulatory fixed cost R_j is not observed by industry and year; we identify the contest from the bills that set it. Second, firms take both sides, and the model determines each firm's side; we use the sides firms state in their reports. Third, many industries lobby each bill (the median proposal is lobbied by superstars of two industries, and about four in ten by superstars of one); we

Table 8. Tax and Procurement: Moments and Identification

	Tax	Procurement	Identifies
What lobbying buys	book tax rate −1.78 pp (0.67) after a year of lobbying	contracts +0.135 log points (0.067)	κ_T, κ_S
No lobbying	superstars' effective rate 0.272	contracts / sales 0.022; elasticity to sales 0.76	τ_0; g_0, κ_g
Who lobbies	by rank 0.27 / 0.17 / 0.13 / 0.12; 2% to 72% across profit deciles	by rank 0.19 / 0.11 / 0.08 / 0.07; 2% to 59% across contract deciles	μ_C, σ_C
How long	83% still lobbying a year later	75%	λ_C
How much	spending 0.019% of the stake	0.17%	ϕ_C

Notes: Payoffs: regressions of the three-year book tax rate and of log prime contracts on last year's share of quarters lobbied on the front, with firm and year fixed effects, superstar firm-years 2008–2022; standard errors in parentheses. Stake: pretax income (tax) and prime contracts (procurement). All ten deciles and quarterly and annual stay and join rates are targeted.

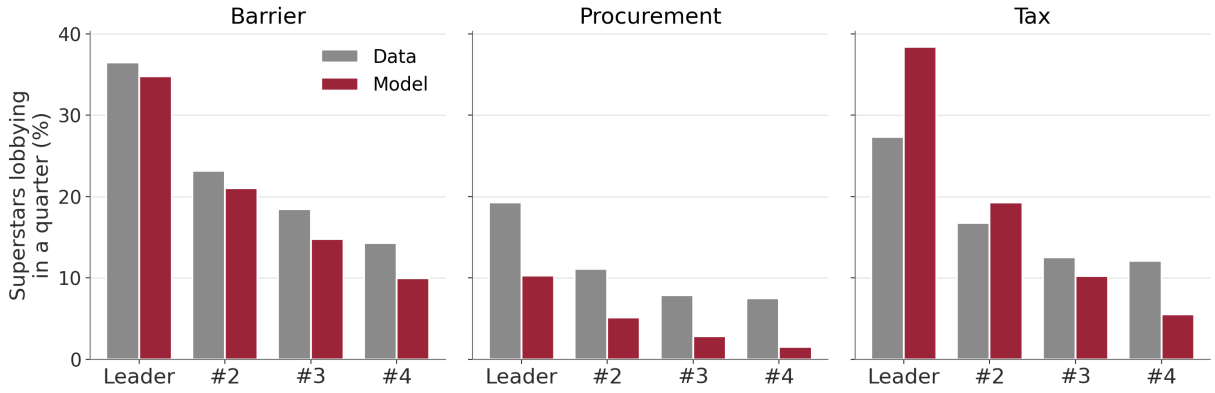


Figure 7. Who Lobbies: Superstars' Participation by Rank, Model and Data
Notes: Share of superstars lobbying on each front in a quarter, by rank within the industry. Data: Compustat superstars, 2008–2022. Model: simulated panel at the estimate.

simulate each real proposal with every industry that lobbied it. Fourth, fringe firms lobby through associations; we use the census of industry associations: their activity, dues, and sides. The block has thirteen parameters: the compliance level c, which sets R_0; the superstars' compliance multiple κ_c; the contest curvature θ and the effectiveness of the anti side β^a; the price of influence κ_R; the participation cost of superstars $(\mu_R, \sigma_R, \lambda_R)$; the association's organizing cost $(\mu_A, \sigma_A, \lambda_A)$, its lobbying cost κ_A, and the coordination cost ϕ_c.

Participation, sides, and the association. Participation of superstars by rank (37, 23, 18, and 14 percent of the first to fourth firms lobby on barriers in a quarter) identifies the mean and dispersion of their participation cost, and its persistence (92 percent continue the next quarter) the redraw rate. The share of stated sides that ask for higher compliance costs, 57.5 percent among superstars, identifies the compliance multiple κ_c: the larger a superstar's own compliance bill relative to a fringe firm's, the more its own cost weighs against the Cournot and deterrence

gains, and the more the smaller superstars ask for lower costs. The elasticity of lobbying spending to sales (0.83) and the leader's share of superstars' spending (0.50) discipline how fast influence rises with the stake, and hence the curvature θ, together with the bills; total lobbying relative to industry sales disciplines the price κ_R. The level of compliance costs, 0.58 percent of output in the measure of Trebbi et al. (2023), identifies c; their broader measure, 1.64 percent, enters as an upper bound. The association's activity (active in 57 percent of quarters, 95 percent of active associations still active the next quarter) identifies its organizing cost; its spending relative to industry sales (2.5×10^{-5}) and the 2.5 percentage point higher participation of superstars when the association is active identify its strength; and the elasticity of dues to the number of fringe firms in the industry (0.35) identifies the coordination cost ϕ_c.

Bills. The enactment of bills identifies how lobbying force translates into regulation. A *proposal* is a set of bills of one Congress that are the same legislation; in 2008–2022, 2,020 proposals raise regulatory fixed costs and 636 lower them. Following Kang (2016), proposal k passes with probability

$$p_k \;=\; \frac{\pi_k + x_k}{1 + x_k + y_k}, \qquad x_k = \sum_{i \in \text{pushers}} \beta^{s_i}\, b_{Ri}^{\theta}\, \omega_{ik}, \qquad y_k = \sum_{i \in \text{blockers}} \beta^{s_i}\, b_{Ri}^{\theta}\, \omega_{ik}, \tag{47}$$

where π_k is its probability of passing without lobbying, $\beta^{s_i} b_{Ri}^{\theta}$ is firm i's force (β^p on the pro side, β^a on the anti side), and ω_{ik} is the share of its force the firm puts on proposal k, chosen optimally and larger where more is at stake. On a raising proposal, pro firms push and anti firms block; on a lowering proposal, the roles flip. Enacted proposals add to or subtract from the stock of rules, which lapse at a constant rate, and Appendix E shows that in the steady state the stock reproduces the contest (13) to first order: at the estimate, lobbying raises the regulatory cost by 22.0 percent according to the bills and by 22.3 percent according to the contest, in the 88 percent of industries in which someone lobbies. We simulate each of the 2,656 real proposals: every industry that lobbied it in the data is drawn from the model's stationary distribution conditional on one of its firms lobbying, forces add up across industries, and the baseline probability π_k depends on the proposal's observable features (sponsor, cosponsors, whether it is must-pass legislation, and Congress) and an unobserved component, fitted at each parameter point. We then run the data's enactment regressions on the simulated proposals.

Table 9 summarizes what the bill moments identify. Proposals that no modeled firm lobbied rarely pass (3 of 267), which pins down the baseline probability. Each log point of lobbying superstars raises the enactment of raising proposals by 5.1 percentage points, which identifies the price of influence κ_R: cheaper influence means more force per firm. The effect does not grow with the size of the lobbying firms: force grows with size as size$^{\varrho}$, with $\varrho = e\theta/(1-\theta)$ where e is the elasticity of lobbying to size, and the data give $\varrho = 0$ (95 percent confidence set -0.35 to 0.40), which disciplines the curvature θ. On lowering proposals the roles flip, which identifies the anti side's effectiveness β^a, although weakly: because unlobbied proposals rarely pass, blocking matters only when β^a is large, and a single firm alone on a raising proposal changes its enactment by $f(1-\pi)/(1+f)$ by

Table 9. Bills: Moments and Identification

In the bill data	Identifies	Data	Model
Lobbying superstars (log), raising proposals	price of influence κ_R	+5.1 pp	+3.5 pp
Any lobbying superstar, raising proposals		−2.7 pp	−1.4 pp
Association lobbying, raising proposals	association's strength	−1.4 pp	+0.4 pp
Lobbying superstars (log), lowering proposals	effectiveness β^a	−0.4 pp	+3.1 pp
Association lobbying, lowering proposals	effectiveness β^a	+1.1 pp	+1.8 pp
Lobbying superstars (log), balanced proposals		+2.9 pp	+3.3 pp
change per unit of the direction index		+1.4 pp	−0.1 pp
Size exponent ϱ	curvature θ	0.00	0.31
Enacted, lobbied, no stated side (fits the baseline)	baseline π	6.7%	6.0%
Lobbying firms stating a side (drawn at data rates)		1.5%	1.6%

Notes: Enactment effects per log point of lobbying firms; linear probability regressions with bill features and Congress fixed effects, on the model's industries; the same code on real and simulated proposals. Direction index: (provisions raising − provisions lowering compliance costs) / all provisions. The first eight rows are targeted; the last two are fitted or set to the data.

pushing but only by $\pi\beta^a f/(1+\beta^a f)$ by blocking.

Estimates. Table 10 reports the estimates. Compliance without lobbying is 2.1 percent of output, and a superstar's compliance bill is 38 times a fringe firm's, so compliance weighs far more on a fringe firm relative to its size: a superstar is on average 280 times larger. The contest curvature is $\theta = 0.68$, and lobbying against the cost is about as effective as lobbying for it ($\beta^a = 1.12$). Participation costs are dispersed and redrawn on average every seven years on barriers. The association's coordination cost is steep, $\phi_c = 4.0$: doubling the membership multiplies the cost of a given lobbying effort by sixteen.

Fit. Table 11 compares the main lobbying moments with the data. The model reproduces who lobbies on barriers by rank (0.35, 0.21, 0.15, and 0.10 of the first to fourth superstars, against 0.37, 0.23, 0.18, and 0.14) and the persistence of participation (0.98 against 0.92), and it reproduces two thirds of the effect of lobbying superstars on the enactment of raising proposals (3.5 against 5.1 percentage points per log point) and the sign and size of the association's effect on lowering proposals (1.8 against 1.1). Seventeen of the 62 lobbying moments are within 25 percent of their data values. The model misses on four margins, and we report them as they are. First, every superstar that takes a side asks for higher compliance costs, against 57.5 percent of stated sides in the data: in the model, firms in the same position have the same stake, and at the estimate the stake is positive for every rank, so the model has no counterpart to the heterogeneity of positions among similar firms. Second, compliance costs are 2.4 percent of output, above the narrow measure (0.58 percent) and the broad one (1.64 percent). Third, as discussed above, the gains from tax and procurement lobbying are too small. Fourth, dues fall with the number of fringe firms in the industry (elasticity −0.33), while in the data they rise (0.35). On the bills, the model also gives superstars too large an effect on lowering proposals (3.1 against −0.4 percentage points) and an

Table 10. Lobbying Parameters

Parameter		Estimate
Regulation		
Compliance without lobbying, share of output	c	0.021
Superstars' compliance multiple	κ_c	38.0
Contest curvature	θ	0.68
Effectiveness of anti-side lobbying	β^a	1.12
Price of influence	κ_R	1.77×10^{-3}
Participation cost of superstars: log mean, s.d.	μ_R, σ_R	−7.90, 2.32
Participation cost redraw rate (per year)	λ_R	0.14
Association organizing cost: log mean relative to superstars', s.d.	$\mu_A - \mu_R$, σ_A	−1.53, 1.80
Association cost redraw rate (per year)	λ_A	0.21
Association lobbying cost: spending / industry sales	κ_A	9.4×10^{-5}
Coordination cost	ϕ_c	3.96
Profit tax		
Statutory tax rate	τ_0	0.290
Lobbying cost: leader's tax cut, share of τ_0	κ_T	0.42%
Lobbying cost convexity	ϕ_T	4.8
Participation cost: log mean, s.d.; redraw rate (per year)	μ_T, σ_T; λ_T	−9.90, 1.74; 0.82
Procurement		
Baseline contracts: contracts / sales	g_0	0.024
Decline of contracts with the gap	κ_g	3.30
Lobbying cost: leader's contracts due to lobbying	κ_S	5.2%
Lobbying cost convexity	ϕ_S	24.0
Participation cost: log mean, s.d.; redraw rate (per year)	μ_S, σ_S; λ_S	−9.27, 1.86; 0.47

Notes: Best point of the joint global search. The lobbying costs of the association and of the tax and procurement fronts are searched over, and reported, in the units of the outcomes they produce (association spending relative to industry sales; a lobbying leader's tax cut; the share of a lobbying leader's contracts due to lobbying), which map one-to-one into the structural cost parameters at each point; the baseline contract is reported as contracts over sales.

effect of lobbying force that grows with the size of the lobbying firms ($\varrho = 0.31$ against 0). The second miss matters most for the welfare cost, because the resources absorbed by compliance are one of its components; Section 6.5 shows how the welfare cost varies across parameter values that fit the compliance level better.

Untargeted moments. Table 12 compares moments that the estimation does not target. The model reproduces who lobbies alongside whom in an industry-quarter: the association alone (0.30 against 0.27), superstars alone (0.14 against 0.15), and both (0.45 against 0.33). It reproduces the share of superstars that lobby on no front (0.65 against 0.62) and on barriers or procurement alone, and the share of industries with at most two superstars (0.09 against 0.12). It misses lobbying on several fronts at once: 6 percent of superstars in the data lobby on all three fronts, against 0.4 percent in the model, in which participation costs are independent across fronts. A shared fixed cost of lobbying, captured by the scope discount ι, would close this gap; we set $\iota = 0$ in the baseline.

Table 11. Lobbying Moments: Model and Data

	Data	Model
Regulation: superstars		
Lobbying in a quarter, rank 1 / 2 / 3 / 4	.365 / .232 / .185 / .143	.348 / .211 / .148 / .100
Still lobbying next quarter	0.924	0.977
Starting next quarter	0.026	0.007
Ask for higher compliance costs (stated sides)	0.575	1.000
Spending–sales elasticity	0.83	1.17
Leader's share of superstars' spending	0.50	0.64
Total lobbying / industry sales ($\times 10^{-5}$)	3.7	9.4
Compliance costs / output	0.58%	2.41%
Regulation: association		
Active	0.573	0.733
Still active next quarter	0.948	0.987
Dues elasticity to the number of fringe firms	0.35	−0.33
Association lobbying / industry sales ($\times 10^{-5}$)	2.5	9.1
Superstars lobbying: association active − inactive	0.025	0.044
Tax		
Effective tax rate	0.272	0.290
Tax rate change after a year of lobbying	−0.0178	−0.0005
Spending / stake	0.019%	0.066%
Still lobbying a year later	0.825	0.612
Procurement		
Contracts / sales	0.022	0.024
Elasticity of contracts to sales	0.76	1.25
Log contracts change after a year of lobbying	0.135	0.031
Spending / stake	0.168%	0.203%
Still lobbying a year later	0.752	0.661
Across fronts		
Lobbying on taxes, given lobbying on barriers	0.516	0.238
Lobbying on procurement, given lobbying on barriers	0.301	0.064

Notes: Selected targeted moments; Appendix G lists all of them. Model: simulated panel and stationary distribution at the estimate.

Local identification. To assess which lobbying parameters the non-bill moments pin down at the estimate, we move each of the 26 lobbying, tax, and procurement parameters by 4 percent of its search range, holding the technology fixed, re-solve the equilibrium, and compute how far each parameter can move, with the others re-adjusting, before the objective worsens by one. The participation parameters of the tax and procurement fronts, the leader's tax cut and lobbied contracts, the compliance level, and the association's spending are tightly pinned (within a third of their ranges). The coordination cost ϕ_c, the dispersion of participation costs, the compliance multiple κ_c, and the contest curvature θ are weakly pinned by these moments, and the pairs they confuse are θ and ϕ_c, and κ_c and σ_R. These are the parameters the bill moments are designed to discipline: the size exponent ϱ for θ, and the enactment effects for the price and the effectiveness of influence. A local analysis that includes the bill moments is in progress.

Table 12. Untargeted Moments

	Data	Model
Who lobbies on barriers in an industry-quarter		
Association only	0.273	0.296
Superstars only	0.146	0.144
Both	0.330	0.447
Neither	0.251	0.114
Fronts a superstar lobbies on (share of firm-years)		
None	0.617	0.649
Barriers only	0.151	0.146
Procurement only	0.025	0.026
All three	0.057	0.004
Industry structure		
Industries with at most two superstars	0.121	0.087
New superstar / leader sales, median (middle #4/#1 tercile)	0.254	0.276
Sides		
Association asks for higher compliance costs	0.358	0.000

6 The Consequences of Lobbying

This section reports the political equilibrium at the estimate, the effect of lobbying on regulation, industry structure, and growth, its welfare cost, and which kind of lobbying causes it. Every comparison is between the estimated economy and the same economy without lobbying: every participation cost set to infinity, so that the regulatory fixed cost is R_0 in every industry and contracts and tax rates are at their baseline, with the same statutory tax rate, procurement budget, and entry cost, and with the fringe pool given by free entry.

6.1 Who lobbies, on which side, and why

Superstars lobby for higher barriers. At the estimate, the stake in regulation (27) is positive for every leader, second, and third superstar and for 99 percent of fourth superstars, so superstars lobby on the side of a higher regulatory fixed cost; the association lobbies against it in every industry state. Figure 8 decomposes the stake by rank, in units of the firm's own after-tax compliance cost. Every superstar gains several times its own compliance cost from a higher barrier, but the source of the gain changes down the ranks. The leader gains 4.8 times its compliance cost from the profits it takes over when marginal fringe firms stop producing, and 3.5 times from the lower rate of fringe breakthroughs: 58 percent of its gross gain is static. For the fourth superstar the shares reverse: it gains 0.9 times its compliance cost from fringe exit and 2.8 times from deterrence, so 75 percent of its gain is dynamic. The net stake falls from 7.3 times the compliance cost for the leader to 3.7 for the second superstar and 2.7 for the third and fourth.

An entrant hurts the weakest superstar most. The deterrence motive of the smaller superstars reflects the asymmetric harm of entry. When a fringe firm breaks into an industry with four



Figure 8. Why Superstars Lobby for a Higher Barrier

Notes: The three components of a superstar's marginal value of a higher regulatory fixed cost (27), by rank, in units of the firm's own after-tax compliance cost, averaged over the stationary distribution of industry states at the estimate. Blue: Cournot gain as marginal fringe firms stop producing. Red: value of fewer fringe breakthroughs. Grey: own compliance cost.

superstars, the entrant ties or passes the fourth firm and the incumbent at the bottom is displaced. In expectation, such an entry lowers the leader's profit by 6 percent, the second's by 8 percent, the third's by 17 percent, and the fourth's by 80 percent (Figure 9), while industry output rises by 1.7 percent. In an industry with a free seat, the entrant takes the seat, no incumbent is displaced, every incumbent loses between 7 and 8 percent of its profit, and output rises by 2.8 percent. The leader's protection from the barrier is mostly protection of its market share; the smaller superstars' is protection of their place.

The fringe opposes the barrier. A fringe firm bears the compliance cost without the protection. By (34), the first two terms of the association's stake (40) sum to the after-tax change in the operating cutoff, which is negative: the survivors' higher profits only partly offset the compliance cost. The third term, the value of a breakthrough to the industry's fringe pool, is too small to change the sign. The association therefore lobbies against the barrier whenever it is organized, which it is in 73 percent of industry-quarters. Its stake is spread over about 5,000 members, and the steep coordination cost ($\phi_c = 4.0$) limits the force it can buy: the superstars win the contest.

The regulatory fixed cost. Averaged over industries, lobbying raises the regulatory fixed cost by 19.6 percent relative to its level without lobbying. The effect varies with the industry's state: the 10th, 50th, and 90th percentiles of R/R_0 across industries are 0.97, 1.20, and 1.52. The barrier is below its level without lobbying in industries in which the association's force exceeds that of the superstars on the regulatory front, typically when few superstars lobby.

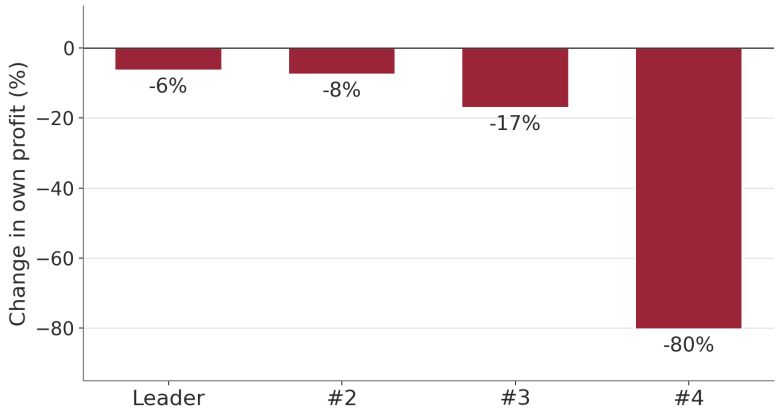


Figure 9. An Entrant Hurts the Weakest Superstar Most

Notes: Expected change in each incumbent's operating profit when a fringe firm breaks into an industry with four superstars, averaged over industry states with the stationary distribution and over landings (10), at the estimate.

6.2 Lobbying, industry structure, and growth

Table 13 compares the economy with and without lobbying. The higher barrier lowers the operating cutoff of the fringe and the expected value of a firm in the fringe pool, so that the pool shrinks under free entry (by 2.7 percent) and a smaller share of it operates: the mass of operating fringe firms falls by 3.5 percent. Fewer operating fringe firms mean fewer breakthroughs into the superstar set, and the stationary distribution of industries shifts toward states with fewer superstars and larger leads: the share of industries with four superstars falls from 74.7 to 73.6 percent, and the shares with one, two, and three rise. The superstars that remain take a larger share of their markets, and their markups rise slightly. Lobbying spending itself is small, 0.024 percent of output; compliance costs rise from 2.09 to 2.41 percent of output.

Composition, not effort. Growth falls by 0.9 basis points, from 1.687 to 1.678 percent. The decline does not come from lower innovation effort by a given firm. In each industry state, superstars innovate slightly *more* with lobbying: a leader's innovation rate rises by 0.2 to 0.3 percent, consistent with the barrier protecting the rents that innovation secures (Figure 10, right panel). Growth falls because the composition of industries changes. Growth is fastest in industries with a close race at the top: where the second firm sells within 10 percent of the leader's sales, the industry's frontier advances at 3.1 percent a year, against 1.2 to 1.4 percent elsewhere. A close race requires a steady supply of challengers, and the barrier reduces it: the share of such industries falls, and the share of industries with a dominant leader rises (Figure 10, left panel). By source, growth from leapfrogging entrants falls the most, from 0.096 to 0.091 percentage points; growth from firms tied with the leader falls from 0.295 to 0.291 points, and growth from the leaders' own innovation is unchanged at 1.296 points. The same pattern holds across the parameter space: among 992 solved points of the search, growth falls at 97 percent of the 412 points at which lobbying raises the barrier by more

Table 13. The Economy With and Without Lobbying

	No lobbying	Lobbying	Change
Regulatory fixed cost, $\mathbb{E}[R]/R_0$	1	1.196	+19.6%
Operating fringe firms per industry	5,017	4,841	−3.5%
Fringe pool per industry	13,998	13,621	−2.7%
Superstars per industry	3.650	3.633	−0.5%
Industries with four superstars	0.747	0.736	−1.1 pp
Leader's share of industry sales	0.0854	0.0865	+1.3%
Compliance costs / output	2.09%	2.41%	+0.31 pp
Fringe fixed costs / output	3.15%	3.01%	−0.14 pp
Lobbying spending / output	0	0.024%	
R&D, superstars / output	0.482%	0.485%	+0.5%
R&D, fringe / output	0.233%	0.228%	−2.1%
Growth rate	1.687%	1.678%	−0.9 bp
Welfare cost (% of lifetime consumption)		**0.89**	

Notes: No lobbying: every participation cost set to infinity, at the same statutory tax rate, procurement budget, and entry cost; the fringe pool solves free entry. Lobbying spending includes participation and organizing costs. Welfare cost: consumption-equivalent variation (45); positive values are losses.

than 2 percent, and rises at 96 percent of the 454 points at which it lowers the barrier by more than 2 percent.

6.3 The welfare cost of lobbying

Lobbying lowers welfare by 0.89 percent of lifetime consumption. Table 14 and Figure 11 decompose the cost with the exact decomposition of Section 4.7. Three findings stand out.

First, most of the cost is static and comes from industry structure. The economy with lobbying has fewer superstars per industry, larger gaps between them, and fewer operating fringe firms. Each of these lowers the output an efficient allocation of labor could produce, and together they account for 0.48 percentage points of consumption, more than half of the total. Misallocation of labor across producers, the margin that markups distort and on which the literature on market power focuses, contributes little: lobbying moves only 0.1 percentage points of labor from the fringe to superstars. Fringe firms employ 87 percent of labor in both economies, against 70 percent in an efficient allocation of labor given the industry states, so the misallocation reflects markups rather than lobbying (Appendix Figure 12).

Second, the resource cost of lobbying is the compliance it buys, not the lobbying itself. Higher compliance costs absorb 0.34 percentage points of consumption, partly offset by the fixed and entry costs that the smaller fringe no longer pays (0.21 points). Lobbying spending, including participation and organizing costs, accounts for 0.03 points. A rent-seeking calculation that counts the dollars spent on lobbying would miss more than 95 percent of its cost.

Third, a quarter of the cost is slower growth. Because a lower growth rate compounds, a fall of 0.9 basis points costs 0.22 percent of consumption. The growth cost operates through the

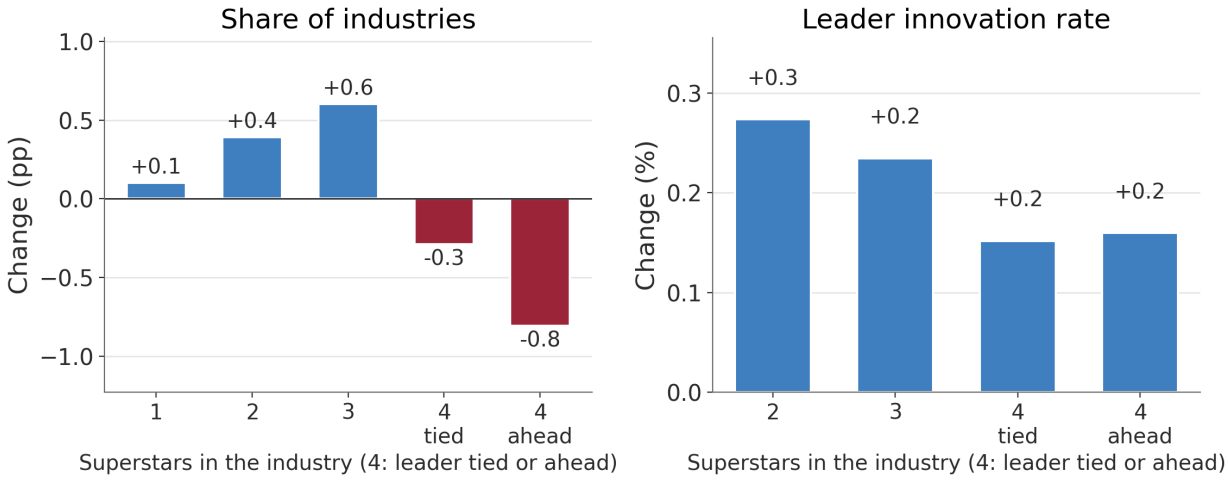


Figure 10. Industry Composition and Innovation With and Without Lobbying

Notes: Left: change in the share of industries by number of superstars (for four superstars, by whether the leader is tied with the second firm or ahead), percentage points. Right: change in the leader's innovation rate in each group of states, percent. Lobbying equilibrium relative to the economy without lobbying, at the estimate.

composition of industries described above, not through the innovation incentives of a given firm.

6.4 Which lobbying causes the cost?

Table 15 shuts down one kind of lobbying at a time, by setting the participation cost of the corresponding front to infinity and solving for the new equilibrium. Barrier lobbying accounts for the entire welfare cost. With lobbying allowed only over regulation, the barrier is 19.6 percent above its level without lobbying and the welfare cost is 0.88 percent of consumption, the same as with all three fronts. With lobbying allowed only for taxes and contracts, the barrier is at its level without lobbying, growth is unchanged, and the welfare cost is 0.0003 percent of consumption. Banning tax lobbying or procurement lobbying alone leaves the welfare cost unchanged.

Why tax and procurement lobbying cost little. In the model, a tax cut won by lobbying is a transfer from the household as taxpayer to the household as owner of the firm, and the government finances it with a slightly higher labor tax. A contract won by lobbying delivers goods to the government at their cost, and the contractor's fee is a transfer; with a fixed procurement budget, the lobbied contract is also a transfer from the other contractors. The only efficiency costs are the lobbying resources and the distortion of labor supply by the labor tax, which rises by about 0.004 percentage points. The distortion has two sides of similar size: the household works less, which lowers output, and enjoys more leisure, which raises utility. In basis points of consumption, lower output costs 0.040, more leisure gains 0.078, and lobbying spending and other margins cost 0.064, for a total of 0.026. Even if the goods bought with lobbied contracts were worth nothing to the household, the cost would rise by only 0.16 basis points.

Table 14. Where the Welfare Cost Comes From

Component	% of consumption	Share
Lower output today		
Industry states: fewer superstars, larger leads	0.299	
Smaller operating fringe	0.177	
Allocative efficiency (labor across producers)	0.029	
Labor input and disutility of labor	0.001	
Total	*0.506*	*57%*
Resources		
Compliance	0.338	
Fringe fixed costs saved	−0.151	
Entry costs saved	−0.057	
Lobbying spending, participation and organizing costs	0.026	
R&D	−0.003	
Total	*0.152*	*17%*
Slower growth		
Total	*0.224*	*25%*
Total	**0.882**	100%

Notes: Exact decomposition of $100\ln(1+\text{CEV})$ (Section 4.7 and Appendix F); positive values are costs of lobbying. The total equals $100\ln(1.00885)$.

Table 15. Which Lobbying Causes the Cost?

Lobbying allowed on	Barrier, $\mathbb{E}[R]/R_0$	Growth	Welfare cost (%)
None	1.000	1.687%	0
Tax and procurement only	1.000	1.687%	0.0003
Barrier only	1.196	1.678%	0.8847
Barrier and procurement (no tax lobbying)	1.196	1.678%	0.8846
Barrier and tax (no procurement lobbying)	1.196	1.678%	0.8855
All three	**1.196**	**1.678%**	**0.8855**

Notes: Each row solves the equilibrium with the participation cost of the excluded fronts set to infinity, at the estimated parameters. Welfare cost: consumption-equivalent variation relative to the economy without lobbying; positive values are losses.

This result inverts the emphasis of the rent-seeking view and of public debate, which focus on firm-specific tax breaks and contracts. These are the most visible products of lobbying, and in the model they are close to pure transfers. The costly product is the least visible one: the contest over the requirements that set the cost of operating in an industry, which changes how many firms compete and how often the leaders are challenged.

6.5 Robustness across the best-fitting points

The estimate is the best point of a global search, and nearby points fit the data almost as well. Table 16 reports the ten points with the lowest objective, together with the best points at which lobbying lowers the barrier.

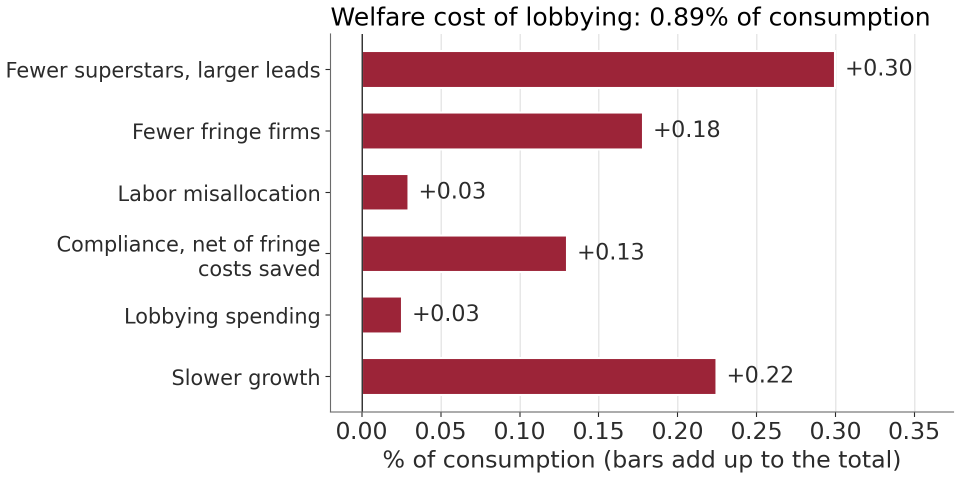


Figure 11. Decomposition of the Welfare Cost of Lobbying

Notes: Components of the welfare cost in percent of lifetime consumption; the bars add up to the total. Compliance is shown net of the fringe fixed costs and entry costs saved; lobbying spending includes participation and organizing costs.

Choice of the estimate. The point with the lowest total objective (52.8) and our estimate (54.3) are within the simulation noise of the objective: re-scoring the estimate with two other seeds of the simulated firm panel gives 53.2 and 53.5. The two points differ sharply in the technology block, however. The first misses where new superstars land by a factor of four (21 percent of its entrants climb two steps above the fourth firm, against 5 percent in the data), and its technology block contributes 13.4 to the objective, against 3.0 at our estimate. Since the welfare cost of lobbying runs through industry structure, entry, and growth, we take as the estimate the point that fits the technology best among the two that fit the data best overall.

The direction of capture is robust; its size is less so. Every one of the ten best-fitting points has lobbying raising the regulatory fixed cost, by between 6 and 150 percent, and every one implies a welfare loss, from 0.03 to 3.7 percent of consumption. The sign of the effect is therefore a robust implication of the data, and of the bills in particular. Without the bill moments, the best-fitting point is one in which the association wins the contest: lobbying lowers the barrier by 36 percent and raises welfare by 0.35 percent. The bill data reject this world. In it, superstars lobby to lower compliance costs, so lobbying superstars would raise the enactment of lowering proposals by 14 percentage points per log point, against -0.4 in the data, and barely raise the enactment of raising proposals (1.6 points, against 5.1); its bill block contributes 73 to the objective, against 9.4 at the estimate. The size of the welfare cost varies more. Across the best points it rises with the compliance burden: the two largest welfare costs (2.5 and 3.7 percent) are at points with compliance costs near 2.4 percent of output, as at our estimate, and the four points closest to the narrow measure of compliance costs (0.3 to 0.7 percent of output) have welfare costs between 0.03 and 0.50 percent. Pinning down the level of compliance costs, which our estimate overstates, is the most important

Table 16. The Best-Fitting Points of the Search

	Objective						
Rank	Total	Technology	Bills	$\mathbb{E}[R]/R_0$	Welfare cost (%)	Δg (bp)	Compliance / output
1	52.8	13.4	10.1	1.24	0.22	−0.4	0.50%
2 (estimate)	**54.3**	**3.0**	**9.4**	**1.20**	**0.89**	**−0.9**	**2.41%**
3	56.5	2.3	8.7	1.77	0.50	−0.4	0.69%
4	57.5	5.8	14.3	1.46	0.90	−0.6	1.41%
5	59.4	5.8	4.7	1.44	0.59	−0.9	0.79%
6	59.6	9.0	1.9	2.50	3.71	−7.7	2.42%
7	61.4	9.1	11.0	1.17	0.34	−0.2	1.13%
8	64.6	3.7	13.8	1.20	0.16	−0.2	0.39%
9	68.9	5.3	2.4	1.83	2.47	−3.4	2.45%
10	72.7	2.8	3.7	1.06	0.03	−0.0	0.29%
Best points at which lobbying lowers the barrier							
11	72.7	3.5	23.8	0.61	−0.56	+0.9	0.37%
51	110.0	6.7	73.0	0.64	−0.35	+0.6	0.27%

Notes: Points of the global search that satisfy the growth constraint, ranked by the total objective. The last row is the point with the lowest objective excluding the bill moments; it ranks far below the others once the bill moments are included. Welfare cost: consumption-equivalent variation relative to the economy without lobbying; negative values are gains. Δg: change in the growth rate.

step toward a more precise estimate of the cost of capture.

6.6 Policy implications

Three implications follow. First, reforms of lobbying should target the contest over regulation rather than lobbying in general: banning barrier lobbying removes the entire welfare cost, while banning tax and procurement lobbying removes none of it. Second, the cost of capture runs through entry and the number of firms that compete, so procedural reforms of industry-wide rule-making, such as requirements to assess the burden of a rule on small firms and on entrants, address the margin on which the cost arises. Third, the outcome of the contest depends on the organization of the side that opposes the barrier. The coordination cost of the fringe's association is steep, and the contest is decided by the few firms whose stakes are concentrated; measures that lower the cost of organizing small firms would shift the contest without restricting political activity. Quantifying these instruments within the model is part of our ongoing work.

7 Conclusion

We measured what firms lobby for, instrument by instrument, and quantified the consequences in a dynamic model of competition and innovation. Firms agree on transfers and disagree on the regulatory fixed costs of their industries, and the largest firms ask most often for higher ones. In the estimated model, superstars lobby for higher barriers, the leaders to take over the business of the marginal small firms and the smaller superstars to keep entrants out; the association of small firms opposes them but cannot overcome the cost of coordinating its members. Lobbying raises the

regulatory fixed cost by 20 percent, makes industries more concentrated, shrinks the competitive fringe, slows growth, and costs 0.89 percent of lifetime consumption. Barrier lobbying accounts for the entire cost, while lobbying for tax cuts and contracts, the most visible and most criticized form of corporate political activity, is close to a pure transfer.

Ongoing work. The welfare analysis compares balanced growth paths; the transition after a ban on barrier lobbying, during which the industry distribution adjusts slowly while lobbying spending stops at once, is the next step. Several parts of the estimation are being refined. The local refinement around the estimate and standard errors are in progress. Three misses of the model point to extensions: the model gives every superstar a positive stake in higher barriers, while 42.5 percent of the sides superstars state in the data ask for lower ones; the compliance burden is above the direct measures; and the model's industries are less concentrated than in the Census. Heterogeneity in how compliance costs scale with size across instruments, which the instrument-level classification can measure, is a natural way to generate opposing positions among large firms. Finally, the instrument-level classification makes it possible to evaluate reforms of individual instruments, for example licensing requirements separately from reporting duties, and to extend the model to the other wedges firms lobby on, such as government demand and administered prices.

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A Static Cournot Equilibrium under Decreasing Returns to Scale

This appendix derives the static product-market objects used in Section 4: the superstar best responses, markups, and profits; the fringe output and profits; and the relative-output system solved at each industry state. The derivation follows Cavenaile et al. (2026), Appendix C.1, adapted to our notation. We fix an industry and a point in time and drop the j and t subscripts throughout. The industry contains N superstars with productivities $\{q_i\}_{i=1}^N$, an active fringe of mass m^{a} with common productivity q_c, the wage is w, and aggregate output is Y. Government contracts do not enter: government purchases are purchases of the final good, and a firm's contracts depend on its gap, not on its output choice.

Demand and notation. The final-good producer's problem yields the inverse demands

$$p_i \;=\; \frac{y_i^{-1/\eta}\,Y}{D}, \qquad p_c \;=\; \frac{\tilde{y}_c^{-1/\eta}\,Y}{D}, \qquad D \;\equiv\; \sum_{k=1}^{N} y_k^{\frac{\eta-1}{\eta}} + \tilde{y}_c^{\frac{\eta-1}{\eta}}. \tag{48}$$

It is convenient to define, for each superstar i and for the fringe aggregate,

$$\Sigma_i \;\equiv\; \sum_{k=1}^{N}\left(\frac{y_k}{y_i}\right)^{\frac{\eta-1}{\eta}} + \left(\frac{\tilde{y}_c}{y_i}\right)^{\frac{\eta-1}{\eta}} \;=\; \frac{D}{y_i^{\frac{\eta-1}{\eta}}}, \qquad \Sigma_{-i} \;\equiv\; \Sigma_i - 1, \qquad \Sigma_c \;\equiv\; \frac{D}{\tilde{y}_c^{\frac{\eta-1}{\eta}}}, \tag{49}$$

where Σ_i includes firm i's own term (equal to one). Revenues then satisfy $p_i y_i = Y/\Sigma_i$ and $p_c\tilde{y}_c = Y/\Sigma_c$: each producer's share of industry expenditure is the reciprocal of its own-normalized CES aggregator, $\sigma_i = 1/\Sigma_i$ and $\sigma_c = 1/\Sigma_c$, with $\sum_i \sigma_i + \sigma_c = 1$.

Superstar best response. Superstar i chooses y_i taking rivals' quantities $\{y_k\}_{k\neq i}$ and the fringe aggregate $\tilde{y}_c$ as given, internalizing the effect of y_i on D. Inverting the production function (4), $l_i = (y_i/q_i)^{1/\alpha}$, so operating profit is

$$\pi_i \;=\; \frac{y_i^{\frac{\eta-1}{\eta}}\,Y}{D} \;-\; w\left(\frac{y_i}{q_i}\right)^{1/\alpha}. \tag{50}$$

The first-order condition equates marginal revenue with marginal cost:

$$\frac{\eta-1}{\eta}\; y_i^{-1/\eta}\,Y\;\frac{D - y_i^{\frac{\eta-1}{\eta}}}{D^2} \;=\; \frac{w}{\alpha}\; y_i^{\frac{1-\alpha}{\alpha}}\; q_i^{-1/\alpha}. \tag{51}$$

The factor $(D - y_i^{(\eta-1)/\eta})/D^2$ on the left is the Cournot internalization: expanding output depresses the industry price index in proportion to the firm's own weight in D. Multiplying both sides of (51) by $(\alpha/w)\, q_i^{1/\alpha}\, y_i^{(\eta-1)/\eta}$ collects all powers of y_i into $y_i^{1/\alpha}$; raising to the power α and substituting (49)

gives the best response

$$y_i \;=\; \alpha^{\alpha}\left(\frac{Y}{w}\right)^{\alpha}\left(\frac{\eta-1}{\eta}\right)^{\alpha} q_i \left[\frac{\Sigma_{-i}}{\Sigma_i^2}\right]^{\alpha}. \tag{52}$$

Markups. From (48) and (49), $p_i = Y/(y_i\Sigma_i)$. Substituting this into the right side of (51) shows that marginal cost equals $\frac{\eta-1}{\eta}\, p_i\, \Sigma_{-i}/\Sigma_i$, so the markup is

$$M_i \;=\; \frac{p_i}{\mathrm{MC}_i} \;=\; \frac{\eta}{\eta-1}\cdot\frac{\Sigma_i}{\Sigma_{-i}} \;=\; \frac{\eta}{(\eta-1)(1-\sigma_i)}. \tag{53}$$

The markup formula as a function of relative outputs is identical to the constant-returns baseline of Cavenaile et al. (2026); decreasing returns alter the level of quantities but not the mapping from market shares to markups.

Superstar profits. The wage bill satisfies $wl_i = w\, y_i^{1/\alpha} q_i^{-1/\alpha} = \alpha\, \mathrm{MC}_i\, y_i$, so using (53),

$$\pi_i \;=\; p_i y_i - w l_i \;=\; \frac{Y}{\Sigma_i}\left[1-\alpha\,\frac{\eta-1}{\eta}\,\frac{\Sigma_{-i}}{\Sigma_i}\right] \;=\; \frac{\big(\eta+[(1-\alpha)\eta+\alpha]\,\Sigma_{-i}\big)\,Y}{\eta\,\Sigma_i^2} \;=\; \sigma_i\Big(1-\frac{\alpha}{M_i}\Big)Y,$$

and the wage bill is $wl_i = \alpha\,\sigma_i Y/M_i$. The term $(1-\alpha)\eta+\alpha$ multiplying Σ_{-i} captures the additional rent that decreasing returns generate on inframarginal units.

Relative outputs. Taking the ratio of the pre-exponentiation form of (51) for two superstars i and k and raising to the power α yields

$$\left(\frac{y_i}{y_k}\right)^{\frac{\eta(1-\alpha)+\alpha}{\eta}} \;=\; \frac{q_i}{q_k}\left[\frac{\Sigma_i - 1}{\Sigma_i - \left(\frac{y_k}{y_i}\right)^{\frac{\eta-1}{\eta}}}\right]^{\alpha}, \tag{54}$$

where both sums are normalized by y_i. Combining the superstar best response (52) with the fringe aggregate output (56) below gives the superstar-to-fringe condition

$$\left(\frac{y_i}{\tilde{y}_c}\right)^{\frac{\eta(1-\alpha)+\alpha}{\eta}} \;=\; \left(\frac{\eta-1}{\eta}\right)^{\alpha}\frac{q_i}{q_c}\,(m^{\mathrm{a}})^{\alpha-1}\left[\frac{\Sigma_{-i}}{\Sigma_i}\right]^{\alpha}. \tag{55}$$

Fringe output and profits. Each fringe firm is a price taker and equates price to marginal cost, $p_c = \frac{w}{\alpha}\, y_{ck}^{\frac{1-\alpha}{\alpha}}\, q_c^{-1/\alpha}$. Using $p_c = Y/(\tilde{y}_c\Sigma_c)$ and $\tilde{y}_c = m^{\mathrm{a}}\, y_{ck}$, all powers of y_{ck} again collect into $y_{ck}^{1/\alpha}$, giving per-firm and aggregate fringe output

$$y_{ck} \;=\; \frac{\alpha^{\alpha}\, Y^{\alpha}\, q_c}{w^{\alpha}\,(m^{\mathrm{a}})^{\alpha}\,\Sigma_c^{\alpha}}, \qquad \tilde{y}_c \;=\; m^{\mathrm{a}}\, y_{ck} \;=\; \frac{(m^{\mathrm{a}})^{1-\alpha}\,\alpha^{\alpha}\, Y^{\alpha}\, q_c}{w^{\alpha}\,\Sigma_c^{\alpha}}. \tag{56}$$

Since $wl_{ck} = \alpha\, p_c\, y_{ck}$, per-firm fringe profit is

$$\pi_{ck} \;=\; (1-\alpha)\, p_c\, y_{ck} \;=\; \frac{(1-\alpha)\, Y}{m^{\mathrm{a}}\, \Sigma_c}. \tag{57}$$

Fringe firms charge no markup, yet decreasing returns ($\alpha < 1$) leave them positive operating profits. This is essential for the lobbying extension: it is these profits, net of the regulatory cost $R(S)$ and the idiosyncratic fixed cost ε, that give the cutoff $\varepsilon^*(S)$ its content.

Solving the static block. Productivities enter only through ratios. By (5) and (6), $q_i/q_c = (1+\lambda)^{-n_i+\bar{a}}/\zeta$, where $\bar{a} = \frac{1}{\bar{N}}\big[\sum_{k=2}^{N} \tilde{n}_{k-1} + (\bar{N}-N)\bar{n}\big]$ is the mean gap of the $\bar{N}$ seats, with empty seats at $\bar{n}$. Given the technology state Θ and the active mass m^{a}, equations (54)–(55) form a system in the N relative outputs $\{y_i/\tilde{y}_c\}_{i=1}^{N}$, which we solve by Newton's method with the analytic Jacobian. Its solution determines the aggregators $\{\Sigma_i, \Sigma_c\}$, hence sales shares, normalized profits π_i/Y and π_c/Y, and markups (53); these depend only on (Θ, m^{a}), not on the wage, which the dynamic solver exploits. The derivatives $\partial(\pi_i/Y)/\partial m^{\mathrm{a}}$ and $\partial(\pi_c/Y)/\partial m^{\mathrm{a}}$ in the stakes (27) and (40) follow from the implicit function theorem on the same system. Labor demands $l_i = (y_i/q_i)^{1/\alpha}$ and $l_{ck} = (y_{ck}/q_c)^{1/\alpha}$ are each proportional to Y/w, and enter only labor-market clearing (42).

B Participation in Lobbying

This appendix gives the objects behind the participation cutoff (25). Fix a superstar with own state s_i in industry state S, and write $e = (S, s_i)$.

Durations. For each front C on which the firm is active, $D_C(e)$ is the expected discounted time over which its current draw is paid, weighted by the scope discount. It solves, on states where $a_{Ci} = 1$,

$$\begin{aligned}\Big(\rho + \sum_{C'} \lambda_{C'} + \sum_{e'} q(e,e')\Big)\, D_C(e) \;&= (1-\iota)^{n_{-C}(e)} \;+\; \sum_{e'} q(e,e')\, D_C(e') \\ &\quad + \sum_{C' \neq C} \lambda_{C'} \Big[\big(1 - G_{C'}(e)\big)\, D_C\big(e^{C',\mathrm{off}}\big) + G_{C'}(e)\, D_C\big(e^{C',\mathrm{on}}\big)\Big],\end{aligned} \tag{58}$$

where $q(e,e')$ are the rates of the events in (22) other than the firm's own redraws (innovation, entry, rivals' and the association's redraws), $G_{C'}(e) = \Phi\big((\ln F^*_{C'}(e) - \mu_{C'})/\sigma_{C'}\big)$ is the probability of being on front C' after its next redraw, and $D_C = 0$ after the firm exits. The firm's own redraw on C ends the spell: it appears in the discount rate with no continuation term.

Values. Since the locked costs enter the flow linearly and do not affect any other choice, the value is affine in them, $v(e) = v^0(e) - \sum_C a_{Ci} F_{Ci} D_C(e)$, where v^0 solves (22) without the locked costs in

the flow and with the redraw terms

$$\lambda_C\Big[\big(1-G_C(e)\big)\,v^0\big(e^{C,\text{off}}\big)+G_C(e)\,v^0\big(e^{C,\text{on}}\big)-D_C\big(e^{C,\text{on}}\big)\,\mathbb{E}\big[F\,\mathbf{1}\{F\le F_C^*(e)\}\big]-v^0(e)\Big]. \tag{59}$$

The expectation $\mathbb{E}[F\,\mathbf{1}\{F \le F^*\}]$ is the partial mean of the lognormal distribution (14) and has a closed form.

Cutoffs and policies. Comparing the values after joining and after staying out, with the new draw subtracted, gives the cutoff (25). The firm's policies (z_i, b_{Ci}) and its stake (27) are computed at its representative locked costs, $\hat{F}_C(e) = \mathbb{E}[F \mid F \le F_C^*(e)]$, so that the locked costs need not be tracked as a continuous state. Resource costs and lobbying spending use the exact distribution of locked costs, from the stationary balance of the locked-cost masses on each front.

C Beliefs about Rivals' Tax and Procurement Statuses

Tracking every superstar's tax and procurement statuses would multiply the industry state space by 4^N. Instead, the state tracks each firm's own statuses, and rivals' statuses are replaced by beliefs. Let $\nu(S, j, b)$ be the stationary measure of superstars at seat j of industry state S with own tax and procurement statuses $b = (a_T, a_S)$. The belief about the statuses of the firm at seat j in technology state Θ is

$$\pi(b \mid \Theta, j) \;=\; \frac{\sum_{S\in\Theta}\nu(S,j,b)}{\sum_{S\in\Theta}\sum_{b'}\nu(S,j,b')}, \tag{60}$$

where $S \in \Theta$ runs over every configuration of the R statuses and the association's status in technology state Θ. Decision objects use the beliefs: rivals' innovation rates and R flip probabilities in (22) are averages of rivals' policies at the actual state over (60), and so are the fringe pool value, the association's cutoff and the contest stakes. The process uses the actual conditional mix, $w(S, j, b) = \nu(S, j, b)/\sum_{b'}\nu(S, j, b')$: the stationary distribution, growth, the locked-cost balances and every moment. Because no participation status enters the conditioning, a firm's participation choice cannot change what it predicts about its rivals. At $\bar{N} = 3$, where the exact model with every status in the state can be solved, the equilibrium with beliefs reproduces growth, the expected number of superstars, the entry flow and the mean regulatory cost to within 5×10^{-6} (relative), and participation on the tax front to within 0.15–1.2%.

D Computing the Equilibrium

At the estimated ladder ($\bar{n} = 5$), there are 84 technology states, 2,180 industry states once the R statuses and the association's status are included, and 33,104 superstar states once each firm's seat and own tax and procurement statuses are included. The equilibrium is computed by three nested fixed points.

1. **Inner**, given the pool m and the contest outcome at every state: superstars' values, dura-

tions, policies and participation cutoffs; the fringe cutoffs, active masses and breakthrough intensities; the association's cutoffs; the beliefs; and the stationary distribution.

2. **Middle**, given m: the stakes from the inner solution, the contest at every state (the regulatory cost, dues and lobbying costs), iterated to a fixed point with Anderson acceleration (Walker and Ni, 2011).

3. **Outer**: the pool m that satisfies free entry (36), by a secant method.

An equilibrium is accepted only if one further evaluation at full precision certifies it: the contest residual is below 10^{-6}, the free-entry residual below 10^{-7} (relative), and the inner fixed point is solved to 10^{-9}. At the estimate, a cold solve takes about twenty minutes on one machine, and the certificate gives a contest residual of 2.2×10^{-7} and a free-entry residual of 2.8×10^{-8}. The economy without lobbying is computed with the same code, with every participation cost set to infinity. The global search of Section 5.1 solves each point in a fresh process with the same certificate; points whose fixed points do not certify within the time limit are discarded.

E From Bills to the Contest

This appendix describes how the enactment of bills maps into the regulatory fixed cost of the contest (13), and how the bill moments are simulated.

Rules add up. The regulatory fixed cost of an industry is the stock of rules in force. Proposals that raise it arrive at rate Λ_U; if enacted, proposal k adds r_k to the stock. Proposals that lower it arrive at rate Λ_L; if enacted, proposal k removes a share s_k of the stock. Rules lapse at rate δ_R. Proposal k is enacted with probability p_k, given by (47), and with probability π_k without lobbying. In the steady state, $\Lambda_U \mathbb{E}_U[rp] = \big(\delta_R + \Lambda_L \mathbb{E}_L[sp]\big)R$, so that

$$\frac{R}{R_0} \;=\; \frac{\mathbb{E}_U[r\,p]}{\mathbb{E}_U[r\,\pi]} \cdot \frac{1+\ell}{1+\ell\,\mathbb{E}_L[s\,p]/\mathbb{E}_L[s\,\pi]}, \qquad \ell \;=\; \frac{\Lambda_L\,\mathbb{E}_L[s\,\pi]}{\delta_R}, \tag{61}$$

where ℓ measures how fast lowering proposals erode the stock relative to lapsing; we fit $\ell = 0.36$. Without lobbying, $p = \pi$ and $R = R_0$. The result is exact when proposals arrive independently of the rules in force and the industry's lobbying is constant while a rule lasts.

How a firm splits its force. Firm i puts a share h_{ik} of its lobbying dollars on proposal k, with mean one across proposals. With the quadratic lobbying cost, its force on proposal k grows as $h_{ik}^{\theta/2}$. To first order in the forces, the value of force on proposal k for R/R_0 is

$$v_k^U \;=\; \frac{r_k}{\lambda_U\,\mathbb{E}_U[r\pi]} \times \begin{cases} 1-\pi_k & \text{push} \\ \pi_k & \text{block} \end{cases}, \qquad v_k^L \;=\; \frac{\ell}{1+\ell}\,\frac{s_k}{\lambda_L\,\mathbb{E}_L[s\pi]} \times \begin{cases} 1-\pi_k & \text{push} \\ \pi_k & \text{block} \end{cases},$$

where λ_U and λ_L are the shares of each type of proposal. Let e_k be the firm's efficiency on the proposal, one for pushing and β^a for blocking. The optimal split is $h_{ik} \propto (v_k e_k)^{1/(1-\theta/2)}$, and we scale the force so that the firm's force on proposal k is $f_{ik} = F_i\, e_k h_{ik}^{\theta/2} / \mathbb{E}[v\, e\, h^{\theta/2}]$, with $F_i = \beta^p b_{Ri}^{\theta}$ on the pro side and $\beta^a b_{Ri}^{\theta}$ on the anti side. Then $\sum_k v_k f_{ik} = F_i$, and to first order the bills give $R/R_0 - 1 = \beta^p B^{\text{pro}} - \beta^a B^{\text{anti}}$, which is the first-order expansion of the contest (13). At the estimate, pro firms put 99.8 percent of their effect on pushing raising proposals and anti firms 97 percent on pushing lowering ones: a proposal that passes 2.5 percent of the time without lobbying is hardly worth blocking. The exact steady state (61) implies that lobbying raises the regulatory cost by 22.0 percent, against 22.3 percent in the contest, in the 88 percent of industries in which someone lobbies; the gap across industry states is 0.2 percent on average and 2.2 percent at most.

Simulating each proposal. We simulate the 2,020 raising and 636 lowering proposals of 2008–2022 as follows. (1) Each proposal is lobbied by the industries that lobbied it in the data. (2) For each such industry, we draw a model industry whose state comes from the stationary distribution, conditional on one of its firms lobbying on barriers. (3) Forces add up across industries; a firm is listed on the proposal if its spending on it passes a threshold, and 1.5 percent of the listed firms state a side, as in the data. (4) The baseline probability is $\pi_k = \Phi(a_{U/L} + c\, z_k + \sigma_\xi\, \xi_k)$, where z_k is the proposal's index in the data's enactment regression (sponsor, cosponsors, must-pass legislation, Congress) and ξ_k an unobserved component; a_U, a_L, c, σ_ξ, and ℓ are fitted at each parameter point to twelve facts of the bill data, including the enactment of proposals that no modeled firm lobbied and of lobbied proposals on which no firm states a side. The sizes r_k and s_k are not observed and are drawn from a lognormal distribution with standard deviation 0.8. (5) The data's own regression code is run on the simulated proposals; industries outside the model's set enter as a control, the number of other lobbyists, as in the data. The twelve facts pin down the baseline only up to the scale of the unobserved component: at the estimate, settings that fit them equally well give a bill block of the objective between 5.6 and 13.0. This range is small relative to the difference between the points at which lobbying raises and lowers the barrier in Table 16. The persistence of enactment chances across re-filed versions of the same proposal measures the scale directly (a correlation of 0.82 between the latent chances of consecutive versions), and we will impose it in the next round of estimation.

Why blocking is weakly identified. A firm with force f alone on a raising proposal changes its enactment probability by $f(1-\pi)/(1+f)$ by pushing and by $-\pi\beta^a f/(1+\beta^a f)$ by blocking. Since π is small, blocking matters only if β^a is large: to first order, lobbying superstars raise enactment on average only if $\beta^a < s(1-\pi)/((1-s)\pi)$, about 19, where s is the pro share. The association's dues fix its blocking force rather than β^a: a larger β^a with less effort gives the same dues and the same blocking. Information on β^a comes from lowering proposals, on which the roles flip, from the direction index, and from the association's effect on lowering proposals.

F Welfare Decomposition

Exact decomposition. Superstar k of an industry produces $y_k = q_k l_k^{\alpha}$, and the fringe of m^{a} identical firms, efficient internally, produces as one producer $Y_c = (m^{\mathrm{a}})^{1-\alpha} q_c L_c^{\alpha}$. Industry output is $y_j = (\sum_k y_k^s + Y_c^s)^{1/s}$ with $s = (\eta - 1)/\eta$, and $\ln Y_0 = \sum_S \mu(S) \ln y(S)$. Since every producer is of degree α in labor, $\ln Y_0 = \alpha \ln L + \ln A^* + \ln E$ exactly, with

$$\ln A^* \;=\; \sum_S \mu(S)\, \frac{1}{e} \ln \sum_k A_k(S)^e, \qquad e = \frac{s}{1 - \alpha s},$$

the productivity of a planner who allocates the same labor equally across industries (all have the same expenditure share and the same α) and within each industry in proportion to A_k^e, given the industry states and the operating fringe. The term $\ln E = \ln Y_0 - \alpha \ln L - \ln A^* \leq 0$ splits into a within-industry and an across-industry part. Welfare (44) is $W = \ln C_0/\rho + g/\rho^2 - \mathcal{D}/\rho$ with $\ln C_0 = \ln Y_0 + \ln(C/Y)$ and $\mathcal{D}$ the disutility of labor, so that

$$100 \ln(1 + \mathrm{CEV}) \;=\; -100 \Big[\alpha\, \Delta \ln L + \Delta \ln A^* + \Delta \ln E + \Delta \ln(C/Y) + \frac{\Delta g}{\rho} - \Delta \mathcal{D} \Big],$$

where Δ is the lobbying economy minus the economy without lobbying. The change in $\ln A^*$ is split into the change due to the industry states and the change due to the operating fringe mass by averaging over the two orders of substitution. The change in C/Y is split by item of the resource constraint (43), and the change in growth by source (41).

Components at the estimate. In percent of lifetime consumption, the components are: labor input -0.031; efficient productivity of the industry states $+0.299$ and of the operating fringe $+0.177$; allocative efficiency within industries $+0.029$ and across industries 0.000; resources $+0.152$; growth $+0.224$; and the disutility of labor $+0.031$; their sum, 0.882, equals $100 \ln(1.00885)$. The resource component splits into compliance ($+0.338$), fringe fixed costs (-0.151), entry costs (-0.057), the association's lobbying ($+0.010$), superstars' barrier lobbying ($+0.006$), the association's organizing costs ($+0.005$), fringe R&D (-0.005), participation costs on barriers ($+0.004$), superstars' R&D ($+0.002$), and participation and lobbying costs on taxes and procurement ($+0.001$). Growth by source falls from 1.6869 to 1.6780 percentage points: leaders' innovation from 1.2961 to 1.2958, innovation by firms tied with the leader from 0.2945 to 0.2911, and leapfrogging entrants from 0.0963 to 0.0910.

G Target Moments

Tables 17 and 18 list every targeted moment, its data value, and its value at the estimate. All moments are weighted equally (Section 5.1).

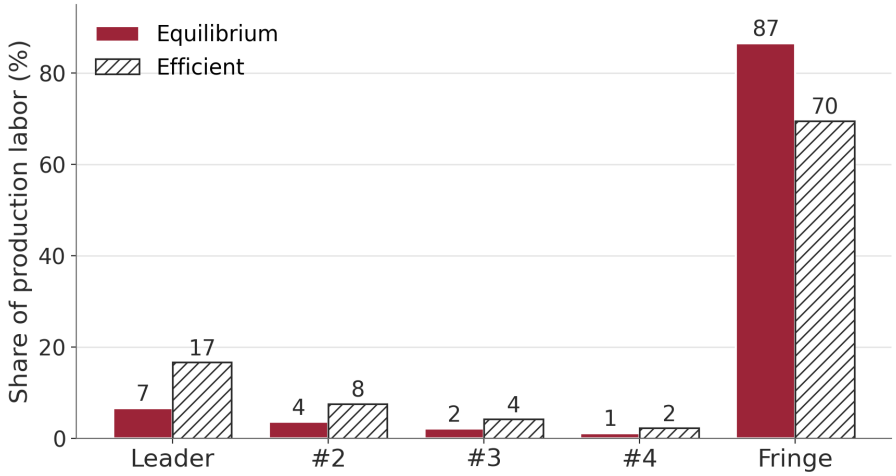


Figure 12. Labor Across Producers: Equilibrium and an Efficient Benchmark

Notes: Shares of aggregate production labor by producer, with and without lobbying, in equilibrium and in the efficient benchmark of Appendix F: a planner who reallocates labor only, holding fixed each industry's state, the operating fringe, and total labor. Within an industry, labor is proportional to A_k^e, $e = s/(1-\alpha s)$, with $A_k = q_k$ for superstars and $(m^{\mathrm{a}})^{1-\alpha} q_c$ for the fringe. Innovation, entry, and fixed costs do not respond.

Technology moments. The panel is Compustat 2008–2022, US-incorporated firms, four-digit NAICS industries outside finance with at least six firms in the year. Superstars are the four largest firms by sales with at least 10 percent of the leader's sales. The bar for a new superstar is the larger of the fourth firm's sales and 10 percent of the leader's, in the year before entry; landing shares count steps of the model's quality ladder in sales (a factor of 1.27 per step). Gap dynamics follow the same pair of firms. Fringe moments use Compustat firms outside the superstar set; R&D is Compustat expenditure on R&D, patents are PatentsView grants matched to Compustat. CR4 is from the Economic Census, entry and exit from the Business Dynamics Statistics, applications from the Business Formation Statistics; growth, R&D intensity, and the labor share are aggregates (Section 2.3). Recession years are excluded from the growth rate.

Table 17. Technology Moments: Data and Model

Moment	Data	Model
Industries with fewer than four superstars	0.250	0.264
Superstars per industry	3.598	3.633
Leader's share of top-four sales, median	0.463	0.480
Leader's share of top-four sales, 90th percentile	0.779	0.691
#2 / #1 sales, 10th percentile	0.219	0.229
#2 / #1 sales, median	0.609	0.671
Industries with #2 / #1 below 0.3	0.180	0.116
Industries with #2 / #1 above 0.8	0.256	0.190

Moment	Data	Model
#4 / #1 sales, median	0.295	0.210
Leader pulls away from a follower within two years	0.0680	0.0419
New superstar's log sales relative to the leader, mean	−1.335	−1.285
New superstar's log sales relative to the leader, s.d.	0.607	0.567
Markup of superstars, cost-weighted	1.263	1.311
Margin of superstars net of R&D	0.304	0.286
Leader's share of top-four R&D	0.489	0.388
Leader's share of R&D, leader–follower pairs	0.636	0.536
Leader's share of patents, leader–follower pairs	0.673	0.506
Fringe share of R&D	0.352	0.320
Fringe firm-years with losses	0.278	0.112
Exit premium of loss-making fringe firms	0.0424	0.0406
CR4 (Economic Census)	0.265	0.174
Growth rate	0.0166	0.0168
Labor share	0.596	0.647
R&D / GDP (business-funded)	0.0197	0.0071
Firm exit rate (BDS)	0.0789	0.0896
Dispersion of log fixed costs, active fringe firms	0.715	0.365
Larger firm's share of patents, pairs far apart	0.777	0.554
Larger firm's share of patents, separated pairs	0.622	0.526
Business applications per firm (BFS)	0.242	0.250
New superstars landing at the bar's level	0.742	0.435
New superstars landing one step above the bar	0.181	0.253
New superstars landing two steps above the bar	0.0531	0.0844
New superstars per industry-year	0.211	0.127
New superstars per industry-year, low #4/#1 tercile	0.151	0.118
New superstars per industry-year, high #4/#1 tercile	0.271	0.155
Applications that become employers within two years (BFS)	0.274	0.359

Table 18. Lobbying and Bill Moments: Data and Model

Moment	Data	Model
Regulation: superstars		
Barriers: superstars lobbying in a quarter, rank 1	0.365	0.348
rank 2	0.232	0.210
rank 3	0.184	0.148
rank 4	0.143	0.0998

Moment	Data	Model
Barriers: still lobbying next quarter	0.924	0.977
Barriers: starting next quarter	0.0260	0.0075
Ask for higher compliance costs (stated sides)	0.575	1.000
Regulation: association		
Association active	0.573	0.733
Association still active next quarter	0.948	0.987
Association starting next quarter	0.0764	0.0369
Dues elasticity to the number of fringe firms	0.346	−0.326
Superstars lobbying: association active − inactive	0.0246	0.0442
Association lobbying / industry sales	2.46×10^{-5}	9.14×10^{-5}
Association lobbying / industry sales (upper bound)	1.83×10^{-4}	9.14×10^{-5}
Regulation: levels		
Barrier spending–sales elasticity	0.832	1.173
Leader's share of superstars' barrier spending	0.497	0.635
Superstars' barrier lobbying / industry sales	3.74×10^{-5}	9.35×10^{-5}
Compliance costs / output	0.0058	0.0241
Compliance costs / output (upper bound)	0.0164	0.0241
Profit tax		
Effective tax rate of superstars	0.272	0.290
Tax rate change after a year of lobbying	−0.0178	-5.01×10^{-4}
Tax lobbying spending / stake	1.89×10^{-4}	6.60×10^{-4}
Tax: superstars lobbying in a quarter, rank 1	0.273	0.384
rank 2	0.168	0.193
rank 3	0.126	0.102
rank 4	0.121	0.0550
Tax: lobbying, pretax-income decile 1	0.0201	0.0381
decile 2	0.0277	0.0642
decile 3	0.0378	0.0830
decile 4	0.0403	0.104
decile 5	0.0781	0.138
decile 6	0.103	0.179
decile 7	0.146	0.232
decile 8	0.280	0.290
decile 9	0.426	0.382
decile 10	0.723	0.499
Tax: still lobbying a year later	0.825	0.612
Tax: starting within a year	0.0474	0.100

Moment	Data	Model
Tax: still lobbying next quarter	0.910	0.879
Tax: starting next quarter	0.0211	0.0310
Procurement		
Contracts / sales	0.0221	0.0242
Elasticity of contracts to sales (PPML)	0.763	1.250
Log contracts change after a year of lobbying	0.135	0.0314
Procurement lobbying spending / stake	0.0017	0.0020
Procurement: superstars lobbying in a quarter, rank 1	0.192	0.103
rank 2	0.111	0.0515
rank 3	0.0788	0.0285
rank 4	0.0746	0.0149
Procurement: lobbying, contracts decile 1	0.0183	0.0059
decile 2	0.0440	0.0099
decile 3	0.0586	0.0128
decile 4	0.0586	0.0348
decile 5	0.0623	0.0484
decile 6	0.0879	0.0379
decile 7	0.132	0.0991
decile 8	0.220	0.0619
decile 9	0.304	0.0601
decile 10	0.590	0.202
Procurement: still lobbying a year later	0.752	0.661
Procurement: starting within a year	0.0340	0.0202
Procurement: still lobbying next quarter	0.885	0.904
Procurement: starting next quarter	0.0154	0.0056
Bills		
Bills: lobbying superstars (log), raising	0.0511	0.0350
Bills: any lobbying superstar, raising	−0.0267	−0.0137
Bills: association lobbying, raising	−0.0137	0.0043
Bills: lobbying superstars (log), lowering	−0.0044	0.0312
Bills: association lobbying, lowering	0.0107	0.0180
Bills: lobbying superstars, balanced proposals	0.0286	0.0332
Bills: change per unit of the direction index	0.0145	−0.0014
Bills: size exponent ϱ	0.00	0.31
Across fronts		
Lobbying on taxes, given lobbying on barriers	0.516	0.238
Lobbying on procurement, given lobbying on barriers	0.301	0.0644

Notes to Table 18: Superstar participation and transitions: Compustat superstars, quarterly reports, 2008–2022; a firm lobbies on a front in a quarter if any of its reports lobbies on it, monitoring included, at high or medium confidence. Association moments: industry trade associations, 2008–2018. Payoff regressions: last year's share of quarters lobbied on the front, firm and year fixed effects. Deciles: of pretax income (tax) and of last year's prime contracts (procurement). Bill coefficients: percentage points per log point of lobbying firms (or per indicator), enactment regressions with bill features and Congress fixed effects on the proposals of the model's industries. Upper bounds enter the objective only when exceeded.

H Issue-Text Splitting: Examples

Each example shows a raw LD-2 issue-text block and the subissues produced by the splitting stage. Deterministic pattern rules handle unambiguous structure (bill-led segments, enumerated lists, blank-line groups); blocks whose boundaries require semantic judgment are routed to the language-model stage. Original filers' typos are preserved.

Example 1: bill-led block, one subissue per bill.

> "S. 1816 and H.R. 3852: Chesapeake Reauthorization, H.R. 5088: America's Commitment to Clean Water Act, H.R. 5509: Chesapeake Bay Program Reauthorization and Improvement Act"

Split into three subissues: (1) S. 1816 / H.R. 3852, Chesapeake Reauthorization (a cross-chamber companion pair kept as one subissue); (2) H.R. 5088, America's Commitment to Clean Water Act; (3) H.R. 5509, Chesapeake Bay Program Reauthorization and Improvement Act.

Example 2: semicolon list mixing instruments.

> "Issues and incentives related to biodiesel, and the Renewable Fuel Standard; S. 1277/H.R. 2238, Biodiesel Tax Incentive Reform and Extension Act; S. 1294, Oil Independence for a Stronger America Act, Title V – Alternative Transportation Fuels; S. 1220, Fulfilling U.S. Energy Leadership Act, Title IIII – Alternative Fuel Deployment. Implementation of Budget Control Act. Extension of biodiesel tax credit. S. 3240, Agriculture Reform, Food and Jobs Act."

Split into seven subissues. The block mixes a regulatory quantity mandate (the Renewable Fuel Standard), a tax-credit companion pair (S. 1277 / H.R. 2238), title-restricted positions on two energy bills, statute implementation, a free-standing tax-extender request, and a farm bill. Keeping it whole would force a single label onto instruments from different wedges.

Example 3: blank-line group with a header.

> "Monitor the following:
>
> Public Law 116-136, Coronavirus Aid, Relief, and Economic Security (CARES) Act, Economic development grant issues; Assistance to Firefighter grant issues; and Municipality related issues

> Public Law 116-260, Consolidated Appropriations Act, 2021, [. . .]
>
> Public Law 116-127, Families First Coronavirus Response Act, Municipality related issues
>
> H.R.6800, Health and Economic Recovery Omnibus Emergency Solutions (HEROES) Act, Municipality related issues"

The header ("Monitor the following:") is distributed to each blank-line segment, and each public law or bill becomes its own subissue with its attached topic list.

Example 4: enumerated list of funding requests.

> "FY 2012 National Defense Authorization and Defense Appropriations Bills and competitive grants seeking funding for the following project initiatives: 1) High Power Laser Technology Initiative.; 2) advanced materials design processing development project; 3) photonics based healthcare technologies for battlefield medicine with the Army; and 4) Comprehensive Solider Fitness. 5) Cancer biomarker detection for early cancer screening."

Split into five subissues, one per numbered item, each inheriting the stem ("FY 2012 NDAA and Defense Appropriations . . . seeking funding for"). All five are labeled appropriations requests, an instrument of the demand wedge.

Example 5: language-model split required (no reliable delimiters).

> "Issues pertaining to intellectual property rights including, but not limited to, patent protection and enforcement, copyright protection and enforcement in international markets, and antitrust, S. 3197 – Platform Competition and Opportunity Act of 2021, S. 2992 – American Innovation and Choice Online Act"

Commas here separate both topics and bills, so pattern rules cannot place boundaries. The language model produces four subissues: (1) patent protection and enforcement; (2) copyright protection and enforcement in international markets; (3) antitrust, S. 3197; (4) antitrust, S. 2992. The intellectual-property topics and the antitrust bills belong to different wedges.

Example 6: title enumeration creates distinct subissues.

> "S 3240, Agriculture Reform, Food, and Jobs Act, and HR 6083, Federal Agriculture Reform and Risk Management Act of 2012 with focus on Titles II, VIII, and IX. HR 5873, Forest Products Fairness Act."

The title enumeration is a real structure: Titles II (conservation), VIII (forestry), and IX (energy) of the farm bills are distinct instruments, so the block yields the per-title subissues plus H.R. 5873 separately.

Example 7: block correctly kept whole.

> "H.R. 1427, Federal Housing Finance Reform Act of 2007. All Provisions. GSE Regulation Reform."

One bill, one position ("All Provisions"), one topic restatement: a single subissue. Internal punctuation alone does not trigger a split; the guards require boundary evidence, and uncertain cases route to the language-model stage rather than being split by rule.

I Policy-Instrument Taxonomy and Model Crosswalk

The policy objects of the model, the industry-wide regulatory fixed cost $R(S)$, the firm's own profit tax τ_i, and its own procurement contracts g_i, must be measured from text that refers to none of them. A lobbying report names a statute, a rule, a bill, or a program; it does not say whether the policy at issue raises a fixed cost of operating or a cost that scales with production. Asking a language model to supply that judgment directly would place the central economic distinction of the paper inside a single step that can be neither inspected nor revised. We therefore separate the reading of the text from its economic interpretation. A closed codebook of policy instruments, each defined by the statutory or regulatory language that identifies it rather than by what it does to costs, is written in advance, and a crosswalk from instruments to the terms of the firm's problem is fixed before any text is classified. The language model assigns text to instruments; the assignment of instruments to economic channels is ours, and takes the form of a fixed table.

I.1 Design

Classification proceeds in three stages. In the first, a language model reads a single text segment and returns, for each distinct request the segment contains, the identifier of a codebook instrument, the conditions defining whom the policy covers, the direction and, where the text states one, the magnitude of the change requested relative to current law, and the position the filer takes. Each of these is a statement about what the text says. In the second stage the identifier is matched to the codebook, which supplies the economics: the wedge the instrument moves, the government account on which it falls, the payoff it typically delivers to a firm that qualifies for it, the base with which that payoff scales, and, for instruments delivered through the tax code, the Treasury provision numbers. The match is mechanical. The third stage derives the quantities the analysis requires: the breadth of the request, whether it favors the firms it covers, the side of the contest on which the filer stands, and the firm's portfolio across wedges.

Two consequences of this division matter. Because the economic content of the classification is held in a table written in advance, it can be examined, disputed, and replaced without reclassifying a single filing, and every result can be recomputed under an alternative crosswalk. Because the classifier's errors and the crosswalk's errors are of distinct kinds, they can also be measured separately. The unit of classification is the provision rather than the bill or the filing: an omnibus statute moves many wedges at once, so a bill enters the panel as a vector of classified provisions, and a subissue containing several distinct requests yields one observation for each. A segment consisting only of a bill reference is recorded as such together with the resolved bill number, and its instruments arrive through the bill's provisions; a segment that names no policy object, or names a policy area too

vaguely to place, is recorded as uninformative and excluded. Neither is forced onto an instrument.

I.2 The codebook and its sources

The codebook is intended to be exhaustive over the policy space on which firms lobby, and its completeness is borrowed rather than asserted: every instrument traces to an official or established inventory of its domain. We draw on the Treasury tax-expenditure inventory, whose provision numbers the codebook retains; the regulatory-forms taxonomy of Xie and Perez (2018); the UNCTAD International Classification of Non-Tariff Measures; the Harmonized Tariff Schedule of the United States and Title 19 of the U.S. Code; the federal Assistance Listings; the Federal Acquisition Regulation; the patent statute and the exclusivity provisions administered by the Food and Drug Administration; the Immigration and Nationality Act; and, as a check on the inventory as a whole, the tools-of-government catalog of Salamon (2002). Coverage failures are not random: they concentrate in the domains a researcher knows least well and bias the measured portfolio toward whatever the codebook covers. Deriving the list from external inventories makes omissions visible to any reader who consults the same sources.

Instruments are organized by what they do to the firm's problem, not by whom they reach. A research credit written for a single firm and one available to the whole economy are the same instrument, distinguished only by the eligibility conditions the classifier reads from the text. Breadth is thus a property of the observation rather than of the codebook, which keeps the observed breadth of lobbying falsifiable.

I.3 Wedges and the crosswalk to the model

A wedge names the term of the firm's problem an instrument moves. Table 19 lists the ten wedges, the margin each moves, and the object of the model to which each corresponds. Three objects of the model are measured from the classification: regulatory fixed costs are the industry-wide burden $R(S)$ of the contest (13); the profit-tax wedge is the object of tax lobbying (11); and procurement contracts, an instrument of the demand wedge, are the object of procurement lobbying (12). The other wedges are recorded rather than discarded, which allows the classification to be reused for extensions and for counterfactuals defined on individual instruments.

Instruments whose incidence is conditional on concentration, among them merger control, conduct rules for dominant platforms, and regulation triggered by size thresholds, are kept separate from R_j, since their burden depends on a firm's position in the industry rather than falling on every active firm alike; they are the natural object of an extension in which lobbying acts on the transitions of the industry state directly. A requirement enters R_j only if its compliance cost is quasi-fixed; any component of the same statute whose cost scales with output is assigned to the output wedge.

Table 19. Wedges and the Crosswalk to the Model

Wedge	Margin moved	Model object
R_j	Industry-wide quasi-fixed cost of operating, entering, or introducing a product	$R(S)$: **barrier lobbying**
b_T	Profit-tax level; scales after-tax profit with no first-order effect on marginal prices	τ_i: **tax lobbying**
s^d	Demand facing the firm: government purchases, buyer subsidies, trade protection	g_i (procurement contracts): **procurement lobbying**; other instruments recorded
s^y	Per-unit wedge on marginal cost or marginal revenue	recorded; extension
s^r	Price of R&D and the private return to innovation	R&D cost χz^{ϕ}; extension
s^k	User cost of capital, including financing	recorded; no capital in the model
s^w	Effective wage bill	labor cost; extension
s^p	Price the firm may charge	markup; extension
$A(\Theta)$	Concentration-conditional rules: act on the industry state or switch on above a size threshold	transitions of Θ; extension
s^i	Lump-sum transfer to, or fee from, the firm	profit level; extension

I.4 What the classifier records

Table 20 lists what the classifier records, one observation for each request it identifies.

Eligibility records only the conditions the text adds beyond the instrument's own definition, and because it is recorded in standard classification systems it also serves as the key that joins an observation to the firm panel. Conditions that bind on quantities the firm chooses are recorded against a closed list of variables that map into the model: output, revenue, price, employment, wages, capital, investment, research spending, profit, market share and concentration, merger events, and firm age. The direction of the requested change is one of a small set of operations, distinguishing the creation of an instrument from its expansion, reduction, extension, or repeal; magnitudes are recorded when the text states them and never imputed. Positions are recorded as the filer states them, and a filing that reports monitoring an issue is kept distinct from one that states no position.

I.5 From classified text to model objects

Besides the wedge, each instrument carries attributes that attach to every observation assigned to it: the government account (expenditure, receipt, or a mandate whose cost is borne privately), the sign of the payoff to a qualifying firm, and its shape, which records whether the payoff scales with a flow or is a lump independent of scale. The shape enforces the rule that only quasi-fixed burdens are barriers.

The scope of a request is determined by the width of its eligibility conditions: a named entity yields a firm-specific request, an industry code with qualifiers a narrow one, an industry branch an industry-wide one, and unrestricted eligibility an economy-wide one. Whether a request favors the firms it covers follows from its direction and the instrument's sign. Let $\xi \in \{-1, 0, +1\}$ denote the

Table 20. What the Classifier Records for Each Request

Field	Content
Instrument	The codebook instrument the text concerns, or a label marking the segment as a bare bill reference or as uninformative
Legislation	Bill and public-law numbers named in the text, resolved to the Congress in which they were introduced
Granularity	Whether the text is finer than the nearest instrument and, if so, a verbatim statement of the finer target
Eligibility	Whom the policy covers, over and above what the instrument itself implies: industry, geography, product, occupation, legal form, and certified status, recorded in standard classification systems
Conditions	Thresholds the policy places on quantities the firm chooses, expressed as comparisons against a closed list of model variables
Effect	The direction of the change requested relative to current law and, when the text states them, its magnitude and starting level
Position	The position the filer takes: support, opposition, either with an amendment, the pursuit of a specific change, monitoring, or none stated
Confidence	The classifier's confidence in the instrument assignment; low-confidence observations are dropped

requested change in the level of instrument k, with creation, increase, extension, reauthorization, and funding taking the value $+1$, reduction and repeal -1, and restatements and studies 0, and let $\text{sgn}_k \in \{-1, +1\}$ be the instrument's sign. Favorability is

$$\text{fav} \;=\; \xi \cdot \text{sgn}_k. \tag{62}$$

The same quantities place an observation on one side of the contest. For an observation on a barrier instrument, let $\text{st} = +1$ when the filer supports the request and $\text{st} = -1$ when it opposes it. The filer would move the regulatory fixed cost in the direction $\text{st} \cdot \xi$: supporting a bill that raises a compliance requirement and opposing the repeal of one are the same political act, and the classification records them on the same side without a further judgment. The share of a firm-year's barrier requests with $\text{st} \cdot \xi = +1$ determines its side, and the share of sides that ask for a higher barrier is the moment that disciplines the superstars' compliance multiple in Section 5.5.

For a few instruments the sign is not a property of the instrument: a tariff protects the firms that produce the good and raises the costs of the firms that use it. Observations on such instruments are resolved filer by filer, first from the stated position, because supporting a duty reveals the protected side, and then from a match between the affected good and the firm's own products or inputs. Classified observations collapse to a panel of firm, quarter, instrument, wedge, and side, from which every moment of Sections 3 and 5 is computed.

I.6 The instruments

The list below groups the 87 instruments of the codebook by wedge.

Regulatory fixed costs *(wedge R_j, 18 instruments)*

- Licenses and charters required to operate; product certification and inspection; pre-market product approval; facility permitting and certificates of need.
- Recurring reporting, recordkeeping, and registration duties; labeling and warning mandates; securities disclosure and shareholder-rights rules; data privacy and cybersecurity rules; one-time outlays required by a standard; financial assurance (surety) bonds.
- Health, safety, and technical requirements on imports; services trade and data-flow restrictions; foreign investment screening; issuance of limited rights to public resources.
- Liability immunities and shields, limits on damages, expansion of liability exposure; conduct rules on how a transaction is made.

Profit tax *(wedge b_T, 7 instruments)*

- The general corporate income tax; deductions from taxable profit; reduced rates on classes of income; reduced tax on foreign and export income; exemption by legal form; deferral and basis rules; deductibility of losses and forgiven debt.

Demand *(wedge s^d, 14 instruments)*

- Government demand: program funding, appropriations, and authorizations; procurement contracts; public infrastructure and services used by firms; tax exemption for state and local government bonds.
- Buyer purchase subsidies and purchase mandates; domestic-content requirements for public purchases; markets reserved by law to designated sellers; advertising and marketing restrictions; government provision of competing services.
- Trade: import tariffs and tariff-rate quotas; limits on import quantities; antidumping and countervailing duties; duty-free trade under preference programs; export controls and sanctions.

Output *(wedge s^y, 17 instruments)*

- Per-unit production credits and support payments; percentage depletion; excise and sales taxes; taxes on emissions and discharges; regulatory levies on revenue; public-resource royalties.
- Technology and performance standards with per-unit costs; bank capital and liquidity requirements; limits on domestic production; tradable permits; portfolio and blending standards; prohibitions of products; obligations to serve; insurance coverage mandates.
- Duty suspensions and drawback; customs procedures and rules of origin; federal insurance backstops.

Innovation cost *(wedge s^r, 5 instruments)*

- R&D tax credits and expensing; R&D grants and technology transfer; patent, copyright, trademark, and trade-secret law; market exclusivity for approved products; enforcement of intellectual property against foreign infringement.

Capital cost *(wedge s^k, 6 instruments)*

- Investment tax credits and expensing; accelerated and bonus depreciation; tax-exempt private-activity bonds; federal lending and loan guarantees; creditor rights in bankruptcy; deductibility of business interest.

Labor cost *(wedge s^w, 8 instruments)*

– Hiring and wage credits; untaxed employee benefits; tax deferral on retirement contributions; payroll taxes; wage floors, hour rules, and benefit mandates; workplace safety and worker protections; employee status and collective bargaining; restrictions on hiring foreign workers.

Set prices *(wedge s^p, 5 instruments)*

– Government-set payment rates; mandatory discounts, rebates, and negotiated prices; price ceilings; price floors; allowed rates of return and revenue caps.

Competition *(wedge $A(\Theta)$, 4 instruments)*

– Merger control and conduct rules; platform gatekeeper regulation; antitrust exemptions; mandated access and interoperability.

Lump-sum transfers and fees *(wedge s^i, 2 instruments)*

– Grants and direct payments to firms; regulatory user fees.

Place-based bundles *(wedge 1 instrument)*

– Zone regimes that combine an employment credit, a financing subsidy, and a capital-gains preference; decomposed into their components, with the geographic restriction recorded as a condition of eligibility.

I.7 Validation protocol

Since the crosswalk is deterministic, validation concerns the classifier's output alone. The completed checks are reported in Section 2.2: agreement with an independent classification by a second language model, agreement with MapLight's hand-coded positions, and the stability of repeated classification. The protocol in progress adds a hand-coded sample. Both authors code a stratified sample of segments independently, with sampling weighted toward the segments carrying the most lobbying dollars, and adjudicate disagreements in writing; agreement between independent coders bounds what any automated procedure can reach, and separates the fields that are intrinsically difficult, chiefly the filer's position and the direction of the change, from those that are close to mechanical, such as the identification of a named statute. Against this sample we will report precision, recall, and F_1 for the assignment of instruments, by instrument and by wedge, per-field accuracy for eligibility, conditions, direction, and position, and the confusion matrix across wedges, and use it to debias the estimates following Carlson and Dell (2025). The final checks operate on the moments rather than the labels: re-computing the target moments without low-confidence observations, with ambiguous observations reassigned to their runner-up class, and at the level of wedges rather than instruments.

I.8 Examples

Consider first a statutory provision: "Producers of electricity from wind receive a credit of 1.5 cents per kilowatt-hour of electricity produced." The classifier assigns it to per-unit production credits, records wind generation as the finer target, sets eligibility to the electric-power industry nationally, and reads the effect as the creation of a credit of \$0.015 per kilowatt-hour from a base of

zero. The codebook supplies the output wedge and a favorable sign; aggregation supplies a narrow breadth, a change of $+1$, and, through (62), a favorable request.

Consider next a provision granting manufacturers with annual revenue below one million dollars a one-time grant of five thousand dollars. The classifier assigns it to grants and direct payments, sets eligibility to manufacturing, records the revenue threshold as a condition on a variable the firm chooses, and reads the effect as the creation of a one-time payment. Because the threshold is recorded as a condition, the observation implies a notch in the firm's problem at one million dollars rather than a transfer to small firms in general.

Consider finally a filing that reads "Support extension of the renewable electricity production tax credit; seek inclusion of energy storage." This is two requests and therefore two observations: an extension of the existing credit with a supporting position, and an expansion of its qualifying technologies to energy storage, with the pursuit of a specific change as the position. Both carry $\xi = +1$ on a favorably signed instrument, so both are favorable requests: an extension and a widening of scope point in the same direction, and the classification recovers this without a further judgment.